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Series 01 · Mathematics and Artificial Intelligence

COMPLETE SIX-VOLUME MONOGRAPH · VOLUMES I–VI

The Grothendieckian Machine

A Mathematical Universe Architecture for Verifiable and Self-Evolving Artificial Intelligence

RESEARCH ARCHITECTURE EDITION 1

I Foundations
II Modern Synthesis
III Architecture
IV Algorithms
V Modular Machine
VI Self-Evolution

Original six-world mathematical design

English · 2026

GAI-MUA: The Grothendieckian Machine

Complete Six-Volume Monograph

GAI-MUA: The Grothendieckian Machine

A Mathematical Universe Architecture for Verifiable and Self-Evolving Artificial Intelligence

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Series Note

GAI-MUA: The Grothendieckian Machine is a six-volume program in mathematics, formal reasoning, system architecture, and self-evolving mathematical artificial intelligence. Its organizing conviction is that major advances frequently arise not from manipulating isolated formulas more aggressively, but from constructing an ambient world in which the objects, transports, invariants, and local-to-global laws of a problem become visible.

The work therefore has two inseparable obligations. The first is mathematical: to explain schemes, topoi, descent, cohomology, motives, higher structures, modern arithmetic geometry, and their successors with exact hypotheses and source-aware proof lineage. The second is architectural: to express mathematical worlds as typed, versioned, provenance-bearing artifacts that can be searched, compared, formalized, computed with, and evolved without allowing statistical confidence to masquerade as proof.

The series uses a strict epistemic constitution. Established mathematics, structural synthesis, proposed definitions, research programs, conjectures, proof obligations, and disputed claims are explicitly distinguished. A theorem enters the established layer only with a checked statement, complete hypotheses, an exact locator, and a recorded dependency closure. A generated formal artifact is not a proof if the target has changed, assumptions have silently strengthened, or the trusted kernel has not replayed the proof.

The phrase *Grothendieckian Machine* names a research architecture, not a completed universal solver. Its purpose is to make mathematical world construction, descent, comparison, obstruction, and reconstruction explicit enough to support rigorous human and machine collaboration.

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Integrated Preface

Purpose and scope

GAI-MUA: The Grothendieckian Machine – A Mathematical Universe Architecture for Verifiable and Self-Evolving Artificial Intelligence is conceived as one monograph in six internal volumes. It joins the history and substance of structural mathematics with the design of artificial-intelligence systems. Its central questions concern the formation of mathematical worlds, the warrant of claims within and between those worlds, faithful representation of structure in typed data, and computational assistance governed by exact sources, explicit hypotheses, independent verification, and human publication authority.

The phrase *Grothendieckian Machine* names the research architecture developed in this work. It carries no attribution of design or endorsement to Alexander Grothendieck. *Grothendieckian* refers to a family of mathematical operations made especially visible by Grothendieck's work and by verified contributions from collaborators, seminar authors, redactors, editors, critics, and successors: passage from objects to relations and functors; relativity to a base; replacement of an inadequate ambient category by a more expressive one; construction through representability; local-to-global passage by descent; retention of automorphisms by categorical ascent; organization of cohomology through functorial operations and comparison maps; reconstruction from categorical, homotopical, or Galois-theoretic shadows; and unification through universal properties.

The word *machine* denotes a federated system of source registries, typed intermediate representations, mathematical-world descriptions, proof and failure artifacts, bounded algorithms, formal adapters, immutable provenance, independent evaluators, publication processes, and governed evolution loops. Exact finite fragments, architectural definitions, research programs, and open obligations retain separate statuses. Mathematical judgment remains distributed across source scholarship, proof, computation, independent review, and authorized publication.

The complete work consequently has four simultaneous scopes.

First, it is a mathematical monograph. Volumes I and II reconstruct a broad Grothendieckian and post-Grothendieckian substrate: schemes, sites, topoi, descent, etale cohomology, stacks, six-functor formalisms, motives, anabelian reconstruction, higher structures, comparison and weight methods, derived and spectral geometry, almost mathematics, perfectoid spaces, diamonds, v-sheaves, prismatic cohomology, condensed and solid mathematics, and selected arithmetic-moduli strategies. The exposition organizes these subjects by mathematical pressure, world-forming operators, and the exact scope of transport.

Second, it is a study in mathematical method and genealogy. The work asks how a theorem, definition, construction, or research program entered the record; which source unit supports attribution; how seminar direction, expose authorship, redaction, editing, and later reconstruction differ; how formulations change across editions; and how disputed arguments should preserve publication, criticism, response, and reception as distinct events. This source-critical layer forms part of mathematical exactness because hypotheses, editions, authorial units, and reformulations determine the claim.

Third, it is an architecture for mathematical intelligence. Volumes III through V translate mathematical operations into typed system obligations. A mathematical world carries its ambient category, objects and morphisms, topology or locality, logical and universe discipline, invariants, admissible transformations, supported fragments, and failure boundaries. A theorem carries statement identity, hypotheses, dependency closure, source lineage, formal target when available, proof or computation receipts, and epistemic status. A bridge records direction, preservation, loss, unresolved compatibility, and inference limits. A computational result records bounds and replay conditions. Proposal, retrieval, compilation, computation, verification, and publication belong to separately governed roles.

Fourth, it is a research program in governed self-evolution. Volume VI studies improvement in retrieval, representation, planning, proof search, counterexample search, tool use, curricula, module interfaces, and coordination policies. The proposed form is a federation of module-local evolution contexts, independent evaluation routes, immutable root epochs, compatible version bundles, external promotion requests, staged deployment, and rollback readiness. Evolution enters the persistent system as evidence-bearing change under an externally governed constitution.

These four scopes explain the project's scale. The mathematical genealogy of Volumes I and II supplies the concepts and cautions from which Volume III derives its architecture. Volume IV turns that architecture into algorithm contracts and bounded procedures. Volume V assigns those procedures to a modular reference platform, publication route, and authority structure. Volume VI governs adaptation and co-evolution while mathematical truth remains attached to source, proof, and review evidence. The integrated edition is the canonical reading object; volume-level editions are proofing and distribution views of the same work.

The monograph presents a selective mathematical synthesis, a source-critical genealogy, a formal architecture, and a research program. Completion claims are confined to exact local results. Its broader aim is a coherent framework in which mathematical world construction, source lineage, formal representation, algorithmic execution, modular implementation, publication, and controlled evolution can be discussed while preserving the differences among theorem, interpretation, definition, experiment, conjecture, obligation, and dispute.

The Grothendieckian method of changing ambient worlds

Many hard mathematical problems arise because the language of the problem suppresses structures needed to state the right invariant, comparison, or gluing law. A characteristic Grothendieckian response diagnoses that pressure, changes the ambient world in a typed manner, and asks whether previously scattered phenomena become functorial, local, representable, or reconstructible. Responsible use records gain, loss, and transport back to the original question.

This monograph calls that pattern *changing ambient worlds*. It includes several distinct operations with separate hypotheses and effects.

One operation is **relativization**. Classical geometric objects become families over a base, and properties are studied through change of base. The expression $X \to S$ makes the base part of the object's type. A morphism $S' \to S$ produces $X \times_S S'$, while exact theorems govern preservation of properness, smoothness, flatness, constructibility, cohomological finiteness, and related conditions. Relativity makes variance visible and converts parameter dependence into functorial obligations.

A second operation is **representability**. One formulates the functor of solutions and asks whether a scheme, algebraic space, stack, or derived object represents it. The unit of inquiry

becomes a universal mapping property. A represented functor carries canonicality, base-change behavior, and access to structural theorems. Representability hypotheses therefore become explicit gates in machine-facing records.

A third operation is **localization and descent**. A global object is approached through local data, restrictions, overlap comparisons, cocycle conditions, higher compatibility, and an effectivity theorem. The topology, cover, nerve, local category, overlap objects, compatibility maps, and reconstruction criterion are declared. Pairwise agreement, cocycle coherence, and effective reconstruction are successive mathematical conditions.

A fourth operation is **categorical ascent**. A moduli problem that identifies isomorphic objects too early may discard automorphisms and distort quotient geometry. Passing from sets to groupoids, from categories to higher categories, or from sheaves to stacks and higher stacks retains coherence data. Equality, isomorphism, equivalence, homotopy, and coherent higher equivalence remain separate relation types. A diagram may commute strictly in one presentation and through a specified cell in another.

A fifth operation is **derivation and stabilization**. Derived categories, cotangent complexes, derived intersections, stable homotopy constructions, and spectral enhancements make extensions, deformation obstructions, and homotopical information visible. Complexes, homotopy limits, and structured ring spectra retain information lost by underived kernels, cokernels, intersections, or deformation spaces. Truncation, cohomology, and comparison functors provide theorem-bearing returns to classical invariants.

A sixth operation is **comparison through a larger mediating world**. Etale, Betti, de Rham, Hodge, crystalline, prismatic, and other realizations provide distinct views of geometric or arithmetic objects. A comparison theorem supplies a directed map or equivalence under specified coefficients, finiteness assumptions, base fields, completions, or period rings. The monograph treats a family of realizations as a prism whose faces are individual comparison claims. Open faces and commutativity conditions become explicit obligations.

A seventh operation is **completion, almost equivalence, tilting, and change of characteristic**. In p-adic geometry, completion, almost mathematics, perfectoid methods, tilting, pro-etale or v-localization, and prismatic mediation create useful ambient worlds. Almost-zero structure differs from zero; tilting differs from identity; untilts carry context-dependent choices; representability of a v-sheaf requires a theorem. Their power depends on controlled loss and constructed return paths.

An eighth operation is **reconstruction from shadows**. In Galois and anabelian geometry one seeks to recover fields, curves, or other geometric objects from profinite groups, categories of covers, outer actions, or related invariants. The theorem states the recoverable objects, input data, relation strength, and hypotheses. Inner ambiguity, base-field dependence, decomposition and inertia data, and functoriality form part of the result. A machine-readable certificate records the shadow, recovery target, recovery strength, and exact theorem.

These operations share a methodological form. Let W be an initial mathematical world, p a problem or failed judgment in W, and $F : W \to W'$ a proposed world-changing operation. A responsible audit records:

- the domain of F on the objects and morphisms under consideration;
- the hypotheses permitting the passage;
- the structures preserved, reflected, forgotten, or identified;
- the existence criterion for the desired construction in W';
- the conclusion class: exact, bounded, derived, truncated, or heuristic;
- the return functor, realization, descent theorem, or reconstruction map;
- the commuting comparison cells and their categorical height;

- the fate of counterexamples and detecting invariants;
- the relation between success in W' and the original question.

This list gives disciplined content to the rising-sea image. A resistant problem may be surrounded by a more capacious theory until structural relations replace direct assault. Enlargement also carries a loss profile: localization may annihilate torsion, a quotient may identify distinct objects, completion may lose global information, a higher setting may replace strict equality by equivalence, and a finite proxy may omit an obstruction beyond its bound. Valid use therefore makes preservation, loss, scope, and return explicit.

The project-native phrase *Diagrammatic Poetry Thinking* names one interpretive synthesis of this practice; it is presented as the author's term. It describes the disciplined use of diagrams, universal properties, adjunctions, covers, comparison maps, and reconstruction patterns as generators of mathematical questions. *Poetry* points to compression and relational imagination. A good diagram may reveal a missing arrow, a naturality square, an obstruction, or a candidate equivalence. The transition from diagrammatic insight to theorem requires hypotheses, source lineage, proof, and exact relation strength; conjecture requires a precise statement and falsification conditions; computation requires a supported fragment and a certificate or failure type.

The architectural volumes operationalize this method. An obstruction profile records the failed judgment and the structures available or absent. An operator profile states the proposed world change, predicted gains, predicted losses, and inference limits. A world specification records inherited, added, modified, and forgotten structure. A task plan fixes bounds and stop conditions before execution. Local computations or proof searches produce receipts, counterexamples, residuals, unknowns, or timeouts. Recovery and transport are separate stages, each with informative outcomes.

Informative failure is essential. A failed attempt may reveal a false theorem, a missing assumption, a category with insufficient expressivity, a coarse topology, a nonconservative invariant, or an inadequate computational fragment. GAI-MUA treats obstructions, counterexamples, unsupported constructs, noninvertible comparison maps, incoherent overlaps, target mismatches, and resource exhaustion as different outputs. Failure becomes part of the mathematical state.

Changing ambient worlds is a family of source-grounded mathematical techniques abstracted into a research calculus. Its legitimate use depends on typed directionality. One may pass from rings to affine schemes contravariantly, from local data to a global object under an effectivity theorem, from a classical object to a derived enhancement, from characteristic zero to characteristic p through a perfectoid tilt under exact hypotheses, or from a geometric object to a Galois-theoretic shadow. Inverse, reconstruction, descent, and comparison arrows each require their own witnesses.

Worlds, bridges, and obligations as the units of mathematical intelligence

The expression *mathematical world* has a more exact intent here than *context* or *domain*. A world is a structured environment in which expressions are well formed, objects have declared types, morphisms have variance, locality has a specified meaning, constructions have universes, and claims have admissible modes of verification. Schematically, a world descriptor contains

$$W = (\mathcal{C}, \mathcal{O}, \mathcal{M}, \tau, \mathcal{L}, \mathcal{I}, \mathcal{T}, \mathcal{H}, \mathcal{B}),$$

where C records the ambient categorical setting, O and M the object and morphism classes, τ a topology or other locality structure when present, $\mathcal{L}$ the logical and universe discipline, $\mathcal{I}$ the available invariants, $\mathcal{T}$ the admitted transformations, $\mathcal{H}$ the standing hypotheses, and $\mathcal{B}$ the supported or studied boundary. This schematic notation serves as an editorial and architectural definition. It makes the fixed data visible before a statement is interpreted.

The boundary $\mathcal{B}$ is indispensable. A world may be a full mathematical category described in prose, a formally implemented fragment, a finite presentation, a truncated higher structure, or a research proposal with unresolved consistency and coherence obligations. Two artifacts can share vocabulary while inhabiting different boundaries. A finite site has enumerated covers; a bounded complex has a declared range; a finite realization diagram has a finite object set. The descriptor keeps the computational fragment aligned with its actual scope.

A **bridge** is a directed relation between world descriptors: a functor, adjunction, localization, realization, base-change operation, comparison transformation, truncation, completion, interpretation, compiler, or theorem-scoped reconstruction procedure. Its record contains source and target worlds, domain conditions, direction, preservation, loss, available return data, coherence obligations, and evidence status. The profile identifies whether the bridge is a faithful functor, an equivalence, a one-way invariant, an analogy, or a proposed translation.

Preservation and reflection are recorded separately. Preservation carries a property from source to image; reflection recovers a property from the image. Faithfulness, fullness, conservativity, reconstruction, and tensor compatibility are distinct predicates. Localization may make a morphism invertible while erasing a distinction visible in the source. These categorical distinctions keep automated translation at its warranted strength.

Loss can be mathematically productive. Forgetful functors, truncations, quotient constructions, completions, and passage to invariants deliberately discard structure. A useful bridge profile may say that connected components are preserved while higher homotopy is forgotten, that a representation retains an outer action while inner choices are identified, or that finite precision determines a residue class while leaving the exact p-adic value open. Downstream reasoning can then use the result at its actual strength.

Return takes several forms: quasi-inverse, adjoint, equivalence after localization, descent theorem, reconstruction theorem, algebraization theorem, comparison isomorphism, or conservative realization. Each has different force. An adjunction supplies units and counits; equivalence requires further conditions. A descent datum supplies gluing input; effectivity supplies the object. A formal object may algebraize under finiteness or properness hypotheses. A shadow may reconstruct an object up to a specified ambiguity. The theorem-level return is named precisely.

Universal properties provide the preferred compression mechanism. A represented functor, product, limit, localization, completion, or adjoint can be characterized by relations to test objects in an admitted class. Robust compression requires typed existence and uniqueness. The machine architecture stores the test category, Hom variance, universal witnesses, comparison maps, and naturality conditions.

Diagrams function as obligation generators. A square can assert strict commutativity, commutativity through a named 2-cell, homotopy commutativity, a Beck-Chevalley comparison, or an unresolved compatibility. A triangle may encode a unit, counit, factorization, or proposed transport. Every semantically relevant cell has a relation type and status, allowing reviewers and checkers to classify the diagram as evidence, goal, definition, or research sketch.

Obstructions provide the dual unit. A failed lift, nonzero cohomology class, incompatible overlap, missing inverse, nonconservative realization, target mismatch, or countermodel identifies why an attempted construction remains open. Cohomological obstruction classes

require theorem-specific structure. Localized types distinguish resource exhaustion, compiler limits, mathematical counterexamples, and cocycle failures. Each output carries its own consequence.

This world-bridge-obligation triad clarifies several foundational examples. Passing from commutative rings to affine schemes reverses arrows. The anti-equivalence is exact on the affine categories, while gluing into general schemes introduces locality and compatibility data. A Grothendieck topology determines the covering families used by its sheaf topos. Comparison theorems govern when different sites present equivalent topoi, so a machine records the chosen basis and topology.

Stacks demonstrate how categorical height preserves information. The set of isomorphism classes may lose stabilizers, while a groupoid-valued or higher-valued functor retains automorphisms and coherence. A coarse moduli bridge has a loss profile that includes stabilizer information and base-change qualifications. An atlas provides access by a scheme or algebraic space while the stack retains its own identity.

Derived enhancement demonstrates the importance of shadows. A derived intersection has a classical truncation that may forget Tor information. A cotangent complex controls deformation and obstruction phenomena beyond an ordinary module of differentials. Cohomology groups extract invariants from a complex. An algorithm on bounded complexes therefore states whether it computes a derived object, cohomology, a spectral-sequence page, or convergence obligations.

Perfectoid and prismatic methods demonstrate mediated comparison. Tilting creates a deep relation between characteristic zero and characteristic p perfectoid worlds. Period sheaves and comparison complexes mediate among cohomology theories. Prismatic cohomology organizes several specializations, each with its own hypotheses and derived structure. A comparison fan records this plurality more accurately than one undifferentiated symbol.

Condensed mathematics demonstrates reconstruction of an ambient category for topological algebra. Sheaf-theoretic objects on an appropriate test category can improve exactness and functorial behavior. Condensed sets, condensed abelian groups, solid modules, and analytic rings remain distinct levels of structure. The bridge from a classical topological object has a declared scope and essential image.

Anabelian reconstruction demonstrates the difference between an invariant and a recovery certificate. A profinite fundamental group with Galois action may determine a hyperbolic curve under a theorem's hypotheses and up to a stated relation. Beyond that scope it remains an invariant. The machine record includes the reconstruction theorem, object class, base conditions, ambiguity, and source. Embedding proximity supplies a retrieval signal rather than an identity judgment.

These examples explain why the monograph's unit of intelligence is the structured inquiry rather than the isolated answer. A mathematically intelligent action may choose the correct world, identify an inadmissible bridge, expose a missing hypothesis, refine a return claim, find a counterexample to effectivity, or abstain because the supported fragment is too small. Such actions improve the structure of inquiry.

The same viewpoint defines the language model's role. It can propose world descriptors, locate possible bridges, and translate prose into candidate IR. Exact source locators, immutable dependency closure, and inspectable relation strength reside in external artifacts. Generated text enters a larger evidence process coordinated with source systems, typed representations, exact and bounded checkers, failure-aware search, and accountable judgment.

Source-critical mathematical genealogy

The genealogy underlying this work is chronological, claim-local, and version-sensitive. It follows programs, proofs, seminars, redactions, correspondence, manuscript layers, later reconstructions, and disputes as distinct source events. Attribution is attached to a person or credited group, an exact source unit, a date or edition, and a specified contribution. Volume-level fame supplies orientation; theorem-level reuse follows the smallest source unit that supports it.

The earliest phase treated here begins with Grothendieck's work in functional analysis during the late 1940s and 1950s. Topological tensor products, nuclear spaces, locally convex spaces, and related duality questions already display a characteristic movement from isolated calculations toward universal constructions and categories adapted to continuity. This preface gives that phase an **abbreviated treatment**. Volume I, Chapter 6, especially its retrospective twelve-theme dossier, supplies the internal source route; the detailed primary corpus of the functional-analysis papers and thesis lies among the **acknowledged omissions** of the present six-volume architecture. The omission concerns theorem-by-theorem exposition, while the genealogy retains the methodological inheritance.

Grothendieck's 1957 paper "Sur quelques points d'algèbre homologique," conventionally called the Tohoku paper, marks the next source-bounded stage. Its exact publication identity and checked sections appear in the Volume II, Chapter 2 source dossier. The paper organizes abelian categories, exactness, generators, filtered colimits, derived functors, and effaceability in a form that reshaped homological algebra. The monograph gives Tohoku an **abbreviated mathematical treatment** through its derived, sheaf-theoretic, and categorical descendants. A complete commentary on every axiom and theorem remains outside the allocated scope.

The Grothendieck-Riemann-Roch and K-theoretic stage followed in the late 1950s. The Borel-Serre paper "Le théorème de Riemann–Roch" (1958) provides the published exposition and proof presentation of Grothendieck's formulation; Volume II, Chapter 2 records that source, while Volume I, Chapter 6 connects it to the retrospective theme of K-theory, intersections, and functorial Riemann-Roch. This preface treats the episode as a bridge from numerical invariants to functorial classes and pushforwards. The wider development of algebraic K-theory receives an **acknowledged selective treatment** through motives, correspondences, and later interfaces.

EGA, SGA, and FGA form the **full foundational treatment** of the genealogy within this monograph. The published EGA installments construct the language of schemes, relative geometry, morphism classes, coherent cohomology, formal methods, and base change. Their title pages associate the work with Grothendieck and state Jean Dieudonne's redactional collaboration. Volume I, Chapter 1 owns the publication architecture and exact edition discipline.

SGA has a seminar source form. Direction, expose authorship, redaction, later proof replacement, and editorial reconstruction are recorded separately. Volume I, Chapter 2 supplies the seminar chronology and thematic map; theorem use retains the expose author and edition. Michael Artin, Jean-Louis Verdier, Luc Illusie, Pierre Deligne, and other named contributors enter through verified exposes, redactions, proofs, and editorial acts. The source unit determines local credit.

FGA has a dispersed lecture, seminar, and publication history. Volume I, Chapter 3 treats its functorial creation of moduli through Picard, Hilbert, Quot, and related representability

problems. The acronym identifies a source constellation rather than a uniform journal sequence. Original materials, later expositions, and modern representability theorems therefore receive separate records.

The years 1970–1972 introduce an ecological and science-critical turn. Grothendieck left the central institutional pattern of his preceding work and participated in *Survivre*, later *Survivre et Vivre*, whose bulletins addressed militarization, nuclear danger, ecological crisis, scientific responsibility, and the social organization of research. The primary source base consists of the dated bulletins and Grothendieck's signed or attributed interventions; later collections and histories serve as secondary context. This preface offers an **abbreviated treatment** of that turn. Volume I, Chapter 6 supplies the internal dossier for the distinction between dated testimony, mathematical reflection, polemic, and later retrospective framing. A comprehensive political history of the movement belongs to the **acknowledged omissions**.

The 1971 Bielefeld manuscript commonly called *Witches' Kitchen* or *Hexenküche* belongs to this same chronological neighborhood while remaining a separate source object. The accessible witness used for this bounded statement is the April 2024 draft transcription distributed by the Centre for Grothendieckian Studies from a scan of the manuscript. It is a transcription witness rather than a critical or scholarly press edition. The preface uses it solely for source identity and chronology. Mathematical or biographical claims requiring folio-level support remain O until a source dossier records the needed passage and review. This is an **acknowledged source with abbreviated treatment**.

The ecological turn matters to the architecture because it changes the scale at which scientific responsibility is considered. Earlier phases of the genealogy concentrate on mathematical objects, methods, and institutions of publication. The *Survivre et Vivre* materials ask how research choices interact with military funding, industrial systems, ecological risk, professional specialization, and public accountability. These questions belong to the history and ethics of science rather than to the proof of a mathematical theorem, yet they inform the monograph's insistence that capability, authority, consequence, and review remain visible.

Source-bounded treatment requires several distinctions. A dated bulletin can establish the position expressed in that issue. A signed text can establish its author's stated argument. An organizational record can establish participation, publication, or a change of name. Later recollection can illuminate self-understanding. Claims about collective motives, influence, or causal effects require additional evidence. The preface therefore uses the 1970–1972 turn to mark a documented change in public activity and scientific criticism, while Volume I, Chapter 6 carries the finer genre distinctions.

This period also provides a historical counterpoint to the machine constitution. The architecture separates the ability to generate a result from the authority to certify, publish, or deploy it. The ecological writings broaden that separation by asking who bears responsibility for the contexts and consequences of research. The monograph translates that concern into provenance, capability policy, security review, resource accounting, incident records, correction, and external authorization. This translation is a present-day synthesis with status S; it is offered as an interpretive lineage rather than as a historical claim that the software contracts reproduce Grothendieck's political program.

The Bielefeld manuscript occupies a related but distinct documentary position. Its title, date, manuscript status, and transcription history support a chronology of intellectual activity around 1971. Its contents deserve passage-level study in a dedicated dossier. Until that work is integrated, the preface treats the source as an acknowledged witness and resists using its evocative title as a substitute for textual evidence. The O status here identifies source and review work still required.

The chronological proximity among departure from IHES, ecological activism, Bielefeld activity, and later withdrawal from central mathematical institutions invites strong narratives. This monograph adopts a finer graph. Institutional departure, public activism, manuscript production, teaching, mathematical correspondence, and later retrospective interpretation are separate event types. Connections among them are admitted according to dated sources. The graph can represent simultaneous activity and changing emphasis while preserving uncertainty about motive and causation.

This finer chronology also prevents a simple division between an early mathematical period and a later reflective period. Mathematical manuscripts and programs continued after 1970, including *La Longue Marche*, the Faltings letter, *Pursuing Stacks*, *Esquisse*, and *Les Derivateurs*. Ecological and spiritual concerns also continued beyond the initial activist interval. The source record therefore shows overlapping strands rather than a single break. The present preface uses this overlap to explain why mathematical genealogy, institutional history, and philosophical reflection require distinct yet connected dossiers.

The early 1980s reopen a sustained manuscript phase. *La Longue Marche à travers la théorie de Galois* (1981) develops a large Galois-theoretic and anabelian program. Volume I, Chapter 95, §95.1 records the archival object, cotes 140-1 through 140-4, and distinguishes its archive-image pagination from the separate 2026 electronic transcription witness. The preface uses the work for chronology and program identity. Section-level mathematical attribution follows the chapter dossier once exact locators are reviewed.

Grothendieck's 1983 letter to Gerd Faltings is a correspondence source with a precise role. It articulates historical programmatic context for curves, fields, rational points, and section questions. Faltings's role in this specific source event is the recipient of the letter. The letter supplies neither a Faltings theorem nor a proof of the section conjecture. Volume I, Chapter 95, §95.1 and Chapter 99, §99.2 distinguish the French publication object "Brief an G. Faltings," pp. 49–58, from the English translation, pp. 285–293, and retain an O status for passage-level claims pending exact concordance.

Faltings's independent mathematical role belongs to a different source line: arithmetic finiteness and the proof of the Mordell conjecture; Tate-type results for abelian varieties; Hodge-Tate and crystalline comparison; Faltings sites; and almost methods that later entered broader p-adic geometry. Volume II assigns these topics to Chapters 35–44. The correspondence role and the arithmetic theorem role therefore remain distinct.

Pursuing Stacks (1983) records an evolving research process in higher groupoids, weak coherence, homotopy types, test categories, stacks, and the derivator program. Volume I, Chapter 4, §4.2 distinguishes the 1983 authorial layer, the 2015 public typesetting layer, the 2021 extended arXiv witness, and the official 2022 SMF Volume I. Their coverage and pagination differ. Volume I, Chapters 103–110 provide the **full thematic treatment** of the mathematical program, while modern infinity-categorical theorems retain their own later sources.

Esquisse d'un Programme (1984 manuscript; 1997 French publication and separate English translation) develops dessins, the Galois-Teichmuller tower, moduli, and anabelian reconstruction. Volume I, Chapter 95 records the distinct French and English pagination regimes. The monograph gives its anabelian horizon a **full programmatic treatment**, paired with theorem-scoped modern reconstruction results and clearly marked conjectural boundaries.

Recoltes et Semailles arose through writing stages from 1983 to 1986 and later acquired distinct transcription and Gallimard edition identities. Volume I, Chapter 6, §§6.1–6.2 records those stages and witness distinctions. The work serves here as primary evidence for Grothendieck's retrospective mathematical self-interpretation and as positioned testimony

concerning mathematical institutions and transmission. Claims about another person's motive, conduct, or priority receive ◌̸ when genuinely disputed or deliberately unadjudicated; a missing edition concordance or source check receives ○.

Les Derivateurs (1990–1991) is a later and separate manuscript program. Volume I, Chapter 107, §107.1 and its source dossier distinguish the original 1,976-page handwritten object from the edited electronic transcription, whose 22 chapter files total 1,171 PDF pages. The page systems serve different witnesses. This monograph gives derivators a **full mathematical treatment** in Chapter 107 and uses modern published sources for Der1–Der4, strongness, stability, Kan-extension calculus, and comparison theorems. The original manuscript supports bounded historical claims while archival shelfmark and quotation-level folio obligations retain status ○.

The collaborator-successor chronology then branches by mathematical transformation. Serre contributes coherent-sheaf foundations, GAGA, local fields, Galois cohomology, correspondence, and critical testing of general constructions. Artin, Verdier, Illusie, and Deligne contribute through representability, algebraic spaces and stacks, derived formalisms, duality, cotangent complexes, deformation theory, weights, monodromy, and comparison. Each contribution is routed to its exact paper, expose, redaction, or later edition in Volumes I and II.

The motivic and homotopical lineages associated with Morel, Voevodsky, and their collaborators develop transfers, $\mathbb{A}^1$-localization, motivic cohomology, and stabilization. Lurie, Toen, Vezzosi, and other authors develop infinity-categorical, derived, and spectral geometry. These are later mathematical worlds with their own theorem sources, even where they answer pressures visible in Grothendieck's programs.

The p-adic chronology requires a precise split. Bhatt-Morrow-Scholze integral comparison constructs the A_{inf}-cohomological mediator and its specializations, with the primary route treated in Volume II, Chapters 73–74. Bhatt-Scholze prismatic cohomology subsequently develops prisms, prismatic sites, and relative prismatic cohomology, treated in Chapters 75–80. The two stages are related through integral comparison and later conceptual reformulation; authorship, source identity, and theorem scope remain distinct.

Perfectoid spaces, pro-etale geometry, diamonds, and v-sheaves are treated through Scholze's papers and their later developments. Condensed and solid mathematics are treated through the Clausen-Scholze lineage and exact modern sources. Gabber-Ramero almost ring theory receives its own systematic attribution. These modern theories form transformation corridors rather than a single successor doctrine.

Anabelian geometry likewise branches. Uchida, Pop, Tamagawa, Mochizuki, and other authors establish results in different field, curve, and categorical scopes. Section conjectures remain conjectural outside proved cases. IUT receives a dispute-specific evidence graph: publication facts, claimed proof edges, critiques, responses, and dated assessments remain separate records. ◌̸ is reserved for disputed, unsettled, or quarantined claims. Missing locators, proof receipts, comparison checks, and review evidence carry ○.

Arithmetic-moduli strategies combine o-minimal counting, Galois-orbit estimates, Hodge theory, and moduli geometry through work of Pila, Zannier, Tsimerman, and other specifically cited authors. Each edge in a strategy graph retains its theorem source and hypotheses.

Three editorial rules govern this chronology. **Primary-source precedence** assigns origin claims to the primary work and uses secondary scholarship for context, correction, and reception. **Version-sensitive concordance** treats manuscripts, transcriptions, translations, official editions, and later reformulations as connected source objects with separate pagination.

Claim-local attribution credits direction, proof, expose authorship, redaction, correspondence, editing, criticism, and later theorem formation according to verified role.

Compact internal source notes for the detailed historical claims are therefore: Volume I, Chapters 1–4 for EGA, SGA, FGA, and *Pursuing Stacks*; Volume I, Chapter 6, §§6.1, 6.2, and 6.6 for retrospective chronology and the early functional-analysis, Tohoku, and GRR themes; Volume II, Chapter 2 for the checked Tohoku and Borel-Serre source identities; Volume I, Chapter 95, §95.1 for *La Longue Marche*, the Faltings letter, *Esquisse*, and the program-object chronology; Volume I, Chapter 99, §99.2 for the section-conjecture correspondence boundary; and Volume I, Chapter 107, §107.1 and its source dossier for *Les Derivateurs*. The *Witches' Kitchen* mention is limited to manuscript and transcription status pending a dedicated internal dossier. These references supply routes to the evidence without creating new bibliography keys.

Source classes and evidentiary reach

Chronology alone gives an order of events; source class determines what each event can establish. The monograph therefore treats journal articles, treatise installments, seminar exposes, correspondence, manuscripts, transcriptions, retrospective testimony, and modern theorem sources as distinct evidentiary forms.

A journal article supplies a public mathematical object with an author list, publication venue, date, and stable internal organization. The article may support a theorem, definition, proof, correction, or stated program according to its text. Later notation can clarify its meaning, yet the original hypotheses and conclusion remain attached to the article. Tohoku and the Borel-Serre GRR paper enter the chronology through this form.

A treatise installment supplies a larger dependency architecture. EGA's installments build successive strata of language, geometry, cohomology, and local structure. The evidentiary unit can be the title page for publication credit, the introduction for program, or a numbered definition and proposition for mathematics. Treatise-level orientation and theorem-level authority therefore coexist at different scales.

A seminar expose combines a seminar ecology with local authorship. The seminar title and directors identify the programmatic setting; the expose byline and text identify the local public source. Later publication may also record redaction, revision, editorial reconstruction, or replacement proofs. SGA citations in this work preserve these layers so that mathematical direction and local contribution remain simultaneously visible.

FGA materials require a constellation model because their source forms include talks, seminar materials, published papers, later collections, and modern expositions. A representability theorem is cited to the source that states and proves it. The broader FGA label records a research world organized around functors, moduli, and universal problems. This dual representation preserves both theorem precision and program coherence.

Correspondence supports a different claim class. A dated letter can establish that its author communicated specified questions, formulations, or strategic ideas to its recipient at that time. The 1983 Faltings letter therefore documents a programmatic communication from Grothendieck to Faltings. The letter's role in section-conjecture genealogy depends on exact passages, while the recipient's own theorems retain their independent sources. Correspondence is especially valuable for chronology and conceptual transmission when its edition and passage identity are secure.

An authorial manuscript supports the work, program, terminology, and dated mathematical development present in that witness. Archive images, typed transcriptions, translations, and official editions are later manifestations with their own identities. *La Longue Marche*,

Pursuing Stacks, *Witches' Kitchen*, and *Les Derivateurs* illustrate different states of manuscript transmission. Their mathematical use follows the witness actually reviewed.

A transcription improves access while adding an editorial layer. It may normalize typography, resolve abbreviations, mark uncertain readings, divide material into files, or supply electronic pagination. These interventions can be scholarly and indispensable. They also create a page and text system distinct from the manuscript folios. The source passport therefore records witness kind, coverage, editor or transcriber, date, pagination, and known open readings.

An official edition supplies publisher metadata, editorial apparatus, and a stable public object. It may cover only part of a manuscript or organize the text differently from an earlier transcription. The 2022 SMF Volume I of *Pursuing Stacks* and the 2021 extended arXiv witness exemplify this distinction. Each can support claims within its verified coverage; concordance connects them.

Retrospective testimony supplies an author's later account of mathematical genesis, institutions, relationships, and self-understanding. *Recoltes et Semailles* is especially rich in this respect. Its mathematical expositions can illuminate how Grothendieck later organized his themes. Its institutional judgments are positioned testimony. Q applies when a claim is genuinely disputed or intentionally quarantined; O applies when a source locator, corroborating source, edition comparison, or review remains missing.

A modern theorem source may realize part of an earlier program in a changed language. The modern result receives its own author, hypotheses, and proof source. A concordance can then classify the relation as realization, reformulation, strengthening, restriction, parallel development, or unresolved comparison. This avoids retrospective transfer of modern theorem content into an earlier manuscript.

These source classes also shape formalization. A theorem source can feed a normalized claim after locator review. A manuscript program may feed conjectures, definitions, or research questions. A letter may feed a dated genealogy edge. A transcription may feed exact text spans once its witness identity is pinned. A retrospective account may feed S interpretations and Q dispute records. The resulting graph reflects evidentiary reach instead of placing every document into one undifferentiated corpus.

The chronology and the logical dependency graph consequently differ. Tohoku precedes many modern derived frameworks historically, while a modern proof may rely on tools developed later. Early p-adic comparison theorems precede A_{inf}-cohomology historically, while Bhatt-Morrow-Scholze integral comparison can furnish a later logical route to rational consequences in its scope. Prismatic cohomology follows as a distinct Bhatt-Scholze framework that reorganizes integral structures. Recording both orders prevents historical precedence from being mistaken for proof dependence.

The same principle applies to intellectual influence. Temporal sequence, thematic resemblance, direct citation, correspondence, seminar participation, and proved mathematical dependence are separate edge types. A source dossier can establish each through appropriate evidence. This allows the genealogy to represent continuity, divergence, independent rediscovery, and later synthesis with equal precision.

The treatment levels are explicit. EGA, SGA, FGA, schemes, topoi, descent, cohomology, stacks, motives, anabelian programs, higher structures, and derivators receive full thematic treatment. Early functional analysis, Tohoku, GRR, the 1970–1972 ecological turn, *Witches' Kitchen*, and source transmission histories receive abbreviated treatment. Full political history, complete archival collation, comprehensive biography, and theorem-by-theorem editions of

every manuscript are acknowledged omissions. This allocation preserves both scholarly scope and source honesty.

Exact map of the six volumes

Volume I: Foundations of the Grothendieckian Universe

Volume I, *Foundations of the Grothendieckian Universe: Schemes, Topoi, Descent, Cohomology, Motives, and Higher Structures*, contains a 120-chapter mathematical program organized in twelve parts, preceded by its prospectus. It supplies the foundational concepts and the source discipline inherited by every later volume.

The opening source chapters establish the publication architectures of EGA and SGA, the dispersed source world of FGA, the manuscript status of *Pursuing Stacks*, the programmatic role of *Esquisse d'un Programme*, the reflective role of *Recoltes et Semailles*, and the need for exact source versions, translations, errata, and theorem identifiers. This first part prevents the mathematical exposition from treating source-family names as theorem locators.

The second part develops structural method: objects through morphisms; Yoneda-style relational reconstruction; relative mathematics and change of base; universal properties; commutative diagrams as generators of obligations; adjunctions; locality and gluing; the rising-sea strategy and its limits; obstructions as first-class objects; and the passage from diagrammatic insight to theorem, counterexample, or conjecture. These chapters introduce the conceptual grammar used later by the machine architecture, but they do not convert metaphors into established mathematics.

The third part presents schemes and relative geometric worlds. Affine spectra, locally ringed spaces, functors of points, fiber products, morphism classes, flatness, smoothness, etaleness, dimension, coherent and quasi-coherent sheaves, formal schemes, algebraization, approximation, and moduli functors are treated as an interconnected architecture. The central lesson is that geometric properties are typed by bases, morphisms, and stability theorems.

The fourth part turns to Grothendieck topologies, sites, sheaves, topoi, points, internal logic, classifying topoi, comparison lemmas, cohomology in a topos, and the transition toward higher topoi. A topos is not presented as a decorative generalization of a space. It is a variable mathematical universe in which locality, logic, and cohomology can be reorganized.

The fifth part gives descent its own systematic treatment: fibered categories, pseudo-functors, Cech nerves, effective descent, fpqc, fppf, etale and smooth descent, descent of properties and structures, cohomological descent, hypercovers, gerbes, and obstruction to gluing. Descent is then abstracted as a general research method, with explicit attention to the difference between compatibility and effectivity.

The sixth part develops etale topology and cohomology: etale morphisms and sites, finite etale covers, Galois categories, the etale fundamental group, torsors, nonabelian H^1, derived functor cohomology, proper and smooth base change, constructibility, duality, nearby and vanishing cycles, monodromy, comparison, and etale homotopy types.

The seventh part explains why moduli problems require stacks. It proceeds through fibered categories, prestacks, stackification, Deligne-Mumford and Artin stacks, atlases, diagonals, inertia, stabilizers, quotient stacks, cotangent complexes, deformation theory, derived enhancements, higher stacks, truncation, and coherence beyond strict commutativity.

The eighth part organizes the six operations and duality. Pullback, direct image, exceptional operations, tensor product, internal Hom, adjunctions, units, counits, projection formulas, Beck-Chevalley transformations, Verdier duality, perverse sheaves, and extensions to stacks and higher coherent settings are treated as a structured family. The point is not to list six

functors, but to retain their domains, adjunctions, exchange maps, coherence laws, and theorem-specific hypotheses.

The ninth part studies motives and realization worlds: algebraic cycles, correspondences, pure motives, equivalence relations, projectors, standard conjectures, mixed motives, triangulated motives, motivic cohomology, $\mathbb{A}^1$-homotopy, motivic spectra, realization functors, motivic six operations, and incomplete realization prisms.

The tenth part treats anabelian reconstruction, including profinite groups, outer Galois actions, number-field and function-field reconstruction, hyperbolic curves, section conjectures, relative and categorical forms, homotopical refinements, and machine-readable reconstruction certificates. Established theorems and open conjectures remain sharply separated.

The eleventh part reads *Pursuing Stacks* as a research program and compares its higher-groupoid, *n*-stack, test-category, derivator, and higher-descent themes with modern infinity-categorical formulations. The purpose is concordance, not retrospective identity.

The final part synthesizes the foundation into mathematical-world descriptors, typed world changes, descent ledgers, representability gates, six-operation coherence cubes, realization prisms, reconstruction certificates, formal-proof interfaces, and theorem cards. These are proposed research instruments. Their definitions belong to the monograph; the mathematics recorded in them retains its independent source status.

Volume I should be read whenever a later system component uses words such as *world*, *base change*, *descent*, *coherence*, *realization*, or *reconstruction*. Those words have mathematical content before they become software fields.

Volume II: The Modern Grothendieckian Synthesis

Volume II, *The Modern Grothendieckian Synthesis: Higher, Derived, Motivic, Perfectoid, Condensed, and Anabelian Geometry*, also contains a 120-chapter program in twelve parts. Its task is not to narrate a simple succession after Grothendieck. It maps the transformations by which foundational ideas were extended, corrected, specialized, and recomposed.

The first part establishes the collaborator-successor source constellation. It studies the ecology around Grothendieck, the Serre-Grothendieck exchanges, Deligne's transformation of cohomological structure, the contributions of Artin, Verdier, and Illusie, later arithmetic geometry, motivic and homotopical descendants, higher and derived descendants, and p-adic, condensed, anabelian, and arithmetic-moduli lineages.

The second part extracts a transformation grammar: comparison; filtration, weight, and purity; derivation and enhancement; localization and stabilization; completion and almost equivalence; categorical ascent; tilting and characteristic change; reconstruction; counting; and modern forms of the rising-sea method. Each transformation is directional. The chain of modern theories records interfaces, not equivalences.

The third part develops coherent-sheaf foundations, GAGA, local fields, Galois cohomology, Hodge structures, mixed Hodge structures, weights, purity, monodromy, perverse sheaves, decomposition phenomena, Tannakian categories, and weight-aware realization records. It shows how comparison is strengthened by filtrations and weights without becoming an indiscriminate identification of cohomology theories.

The fourth part treats Artin representability, algebraic spaces, stacks, deformation functors, cotangent complexes, obstruction theories, derived categories, derived intersections, and derived moduli. It explains how representability and derivation cooperate to create moduli worlds able to retain infinitesimal and homotopical information.

The fifth part follows arithmetic finiteness, Tate-type results, period rings, Hodge-Tate and de Rham representations, crystalline and semistable comparison, Faltings sites and topoi,

almost zero objects, almost purity, systematic almost ring theory, and return paths to ordinary algebra.

The sixth part develops correspondences, transfers, triangulated motives, motivic cohomology, $\mathbb{A}^1$-homotopy, motivic spectra, norm-residue dependency architecture, realization functors, and the open boundaries of motivic six operations.

The seventh part presents infinity-categories, higher limits and colimits, adjunctions, infinity-groupoids, higher topoi, hyperdescent, presentability, accessible localization, derived algebraic geometry, structured infinity-topoi, spectral algebraic geometry, higher stacks, and formalization boundaries created by higher coherence.

The eighth part moves through nonarchimedean and adic preliminaries to perfectoid fields, algebras, and spaces; tilting and untilting; perfectoid almost purity; pro-etale topology; period sheaves; diamonds; v-sheaves; v-descent; quotients; and transformation records.

The ninth part treats integral p-adic Hodge theory, A_{inf}, delta-rings, prisms, prismatic sites, prismatic cohomology, and comparison fans linking Hodge-Tate, de Rham, crystalline, and etale structures.

The tenth part develops extremally disconnected test objects, condensed sets and abelian groups, exactness in condensed categories, solid modules, derived condensed mathematics, analytic rings, and interfaces with analytic geometry.

The eleventh part separates proved anabelian reconstruction from section problems, Mochizuki's established anabelian work, anabelioids and frobenioids, literal IUT objects and claimed implications, critiques and responses, and arithmetic-moduli strategies involving Shimura varieties, o-minimal counting, Galois orbits, and Andre-Oort-type results. This part contains the monograph's most explicit disputed-claim firewall.

The final part builds modern research instruments: source-audited comparison corridors, invariant registries, almost-exactness profiles, higher-coherence compilations, perfectoid-prismatic dependency maps, condensed completion certificates, anabelian benchmarks, IUT evidence graphs, arithmetic-moduli strategy graphs, primary-source concordances, transformation ledgers, theorem atlases, and gate matrices.

Volume II supplies the transformation operators inherited by the machine. It also supplies a caution: every powerful modern passage comes with hypotheses, direction, categorical height, and a nontrivial boundary of failure.

Volume III: Architecture of the Grothendieckian Machine

Volume III, *Architecture of the Grothendieckian Machine: Categorical Intelligence, Verified Knowledge, and Mathematical Universe Engines*, contains 93 chapters after its prospectus. It converts the mathematical and genealogical material of the first two volumes into a technology-neutral normative architecture. It does not claim that the repository already implements that architecture.

The opening chapters define what it means to pass from mathematical foundations to machine obligations. The mathematical universe, rather than the isolated token sequence, becomes the principal unit of representation. Epistemic status is a machine-enforced type. The rising-sea method becomes an operational but falsifiable workflow. Fast proposal systems, slower reasoning systems, and trusted checkers are separated.

The next architectural layer defines immutable artifact envelopes, foundation and transformation schemas, world specifications, typed maps, variance, base change, assumptions, bounds, supported fragments, proof receipts, failure certificates, replay, registries, and dependency closure. These contracts form the thin waist through which later subsystems communicate.

The Grothendieck Mathematical Universe System, or GMUS, is then developed. Its MUE, GCL, and finite MUF-H components organize geometric, cohomological, representation-theoretic, and finite motivic proxies. The Topos Reasoning Engine divides into TNCA, which proposes or constructs candidate worlds in response to expressivity pressure, and TNRA, which routes and reasons within worlds through localization, bounded materialization, computation, proof search, counterexample search, descent, transport, and externally governed promotion requests. A parallel Grothendieckian Research Architecture, GRA, extracts source-audited operator events and transformation lineages.

The Grothendieckian Mathematical Language System, or GMLS, forms another major layer. It passes from structural motifs to lexical primitives, typed notation, grammar, formal-semantic interpretation, theory reconstruction, axiom management, validation, and versioning. The Grothendieckian Contextual Architecture, GCA, represents contexts, goals, covers, local cognitive sections, overlaps, obstructions, reflection, and universe shifts. The Grothendieckian Descent Reasoning Architecture, GDRA, decomposes tasks, plans covers, dispatches capability-limited workers, checks overlaps, repairs failures, and prepares gluing certificates. AURORA is reserved as a small-kernel research path for worlds, manifestations, identities, witnesses, and non-collapse; it is not granted theorem authority.

Later chapters study the GMUS-GMLS boundary, structure-to-language compilation, language-to-world feedback, translation treaties, deadlock, oscillation, semantic drift, human intervention, and end-to-end research campaigns. The final architectural chapters specify trusted-kernel boundaries, proof-assistant and certified-solver adapters, promotion policy, independent replay, security, resource budgets, observability, reproducible builds, incident and retraction protocols, technology neutrality, a polyglot reference implementation, migration, integration milestones, benchmarks, red-team programs, and the roadmap to Volumes IV through VI.

Volume III is where philosophical language becomes contract language. Every inspiring term is required to acquire inputs, outputs, invariants, failure states, provenance, authority limits, and tests.

Volume IV: Algorithms of Grothendieckian Intelligence

Volume IV, *Algorithms of Grothendieckian Intelligence: Categorical Computation, Formal Reasoning, and Verifier-Grounded Discovery*, contains 118 chapter units in its manifest, arranged in twelve parts. Its algorithm catalogue may contain more algorithm records than chapter units because one chapter can specify or trace multiple algorithms; chapter count and algorithm count are therefore not interchangeable.

The first part defines algorithmization itself: effective fragments, bounds, undecidability, exact and certified procedures, bounded and heuristic methods, correctness, completeness, soundness, abstention, the algorithm contract, and epistemic promotion. The canonical contract records typed inputs and outputs, hypotheses, invariants, complexity, failure modes, evidence, provenance, implementation references, tests, axiom closure, and reproducibility.

The second part defines a foundation intermediate representation: typed mathematical signatures, universe and variance checking, diagram and coherence obligations, source-to-IR compilation, target-fidelity checking, proof receipts, independent replay, counterexamples, and failure records.

The third part gives effective procedures for rings, finite presentations, affine schemes, fiber products, base change, gluing, finite sites, presheaves, finite sheaf checking, Cech constructions, certified descent, refinement, obstruction localization, hypercover obligations, and explicit limits on effective topos computation.

The fourth part treats finite etale algebras, cover registries, finite Galois categories, quotient probes of fundamental groups, groupoid-valued descent, presented stack atlases, diagonals, inertia, stabilizers, six-operation type signatures, adjunction and base-change obligations, projection formulas, duality obligations, and external proof backends.

The fifth part develops finite Nori diagrams, realization compilation, multi-realization fusion, endomorphism constraints, finite universal-category proxies, idempotent extraction, Chow-compatible facades, realization prisms, motivic signature matching, bridge plans, residues, and explicit non-equivalence certificates.

The sixth part treats bounded complexes, spectral-sequence propagation, filtrations, convergence obligations, cotangent and deformation records, finite simplicial constructions, truncated higher coherence, homotopy-coherent transport, and the boundary of infinity-categorical computation.

The seventh part gives bounded interfaces for almost, perfectoid, pro-etale, v-sheaf, diamond, and prismatic computations, always retaining finite-precision and provenance limits.

The eighth part covers finite condensed test sites, condensed sheaf checking, solid and exactness obligations, profinite truncation, outer actions, decomposition and inertia, theorem-scoped anabelian reconstruction, experimental theater and alien-copy compilation, witness validation, literal-IUT formalization, consequence firewalls, and arithmetic-moduli strategy graphs.

The ninth and tenth parts specify the TRE, TNCA, TNRA, GRA, GMLS, GCA, GDRA, DDA, and AURORA algorithm families: campaign compilation, operator retrieval, world routing, localization, gluing, event extraction, cognitive graphs, motif mining, lexicon formation, grammar synthesis, semantic grounding, axiom management, contextual cover algorithms, obstruction handling, descent execution, identity checking, and target-fidelity validation.

The eleventh part addresses type-directed retrieval, proof-state search, tactic planning, lemma discovery, proof repair, counterexample and small-model search, obstruction-driven conjecture generation, novelty and duplicate detection, multi-agent candidate generation, and human-machine adjudication.

The final part specifies benchmark constitution, unit and metamorphic tests, proof replay, end-to-end GMUS-GMLS campaigns, calibration, abstention, failure quality, cost, scaling, reproducibility, and implementation milestones.

Volume IV's governing distinction is between execution and warrant. An algorithm may terminate exactly, produce a certificate, search heuristically, or remain a research procedure. Those classes are not typographic decorations. They determine what downstream conclusions are legal.

Volume V: The Modular Grothendieckian Machine

Volume V, *The Modular Grothendieckian Machine: Verified Kernels, Mathematical Systems, and Reference Implementation*, contains 72 chapters. It assigns the architecture and algorithms to a modular reference platform while preserving their epistemic ceilings.

The first twelve chapters define the constitutional substrate: immutable roots, content-addressed artifacts, proof receipts, target fidelity, axiom auditing, policy and signing roots, promotion, retraction, rollback, schema compatibility, artifact envelopes, world and diagram schemas, epistemic state machines, event and obligation types, and an inert `EvolutionContract`.

Chapters 13 through 18 define the foundation IR and compiler stack, including categorical and diagram compilation, finite-site and sheaf compilation, descent and higher-coherence

compilation, derived and six-operation representations, motivic and anabelian interfaces, normalization, optimization, and certificate emission.

Chapters 19 through 30 cover source acquisition, licensing, checksums, document normalization, entity and notation resolution, definition and theorem extraction, source lineages, quarantine, theorem and concept graphs, dependency graphs, Merkle provenance, indexes, hybrid retrieval, freshness, contradiction, supersession, and retraction.

Chapters 31 through 36 provide adapters for Lean 4, Coq, Isabelle, SAT, SMT, model finders, Sage, GAP, Singular, PARI, computer algebra, verified numerics, exact algebra, sandboxing, and certificate import. An adapter's authority is limited to the predicate checked by its pinned implementation.

Chapters 37 through 54 implement the proposed mathematical-system layer: MUE, finite MUF-H, TRE, TNCA, TNRA, GRA, GMLS, semantic grounding, theory reconstruction, GCA, GDRA, DDA workers, AURORA, GMUS-GMLS integration, theorem-scoped anabelian adapters, enriched shadows, site-guided cycles, theater-mediated comparison, alien copies, witness validation, literal-IUT formalization, and consequence firewalls.

Chapters 55 through 66 supply the platform: durable workflows, event fabrics, transactional metadata, artifact storage, scheduling, caching, budgets, APIs, SDKs, command-line and notebook interfaces, collaboration ports, scoped identity, capability security, sandboxing, observability, benchmark registries, deployment, backup, disaster recovery, supply-chain control, dependency security, and model security.

The final six chapters cover single-source PDF/DOCX publishing, reproducible HTML/CSS mathematical figures, research and release manifests, end-to-end campaigns, the non-self-promotional evolution boundary, and the handoff to Volume VI.

The constitutional root is a seven-role separation among proof-kernel, target-fidelity, provenance, capability-and-policy, signing, promotion, and rollback functions. It is not a superuser process. Candidate producers, validators, signers, promoters, and rollback authorities remain distinct. A module may create a promotion request, but no ordinary module interface may authorize the requested transition.

Volume VI: The Self-Evolving Grothendieckian Machine

Volume VI, *The Self-Evolving Grothendieckian Machine: Reinforcement Learning, Descent-Coherent Co-Evolution, and Recursive Mathematical Discovery*, contains 80 chapters. It replaces the image of one self-improving model with a federation of governed evolution contexts.

The opening chapters define mathematical self-evolution, prohibit self-certification, establish immutable root epochs, classify evolution types and obligations, distinguish failure from uncertainty, and separate five adaptation timescales.

Chapters 7 through 20 define categorical state, versioned worlds, mathematical partially observable decision processes, observations, beliefs, typed actions, policy morphisms, invariant and obstruction objects, and module-evolution contexts for ingestion, ontology, IR, compilers, MUE, GCL, MUF-H, TRE, TNCA, TNRA, GRA, GMLS, GCA, GDRA, DDA, AURORA, anabelian services, retrieval, proof, and publication.

Chapters 21 through 34 develop hierarchical reinforcement learning, model-based planning, multi-agent policies, tool routing, theorem-proving and proof-repair learning, counterexample and conjecture policies, safe exploration, reward constitution, proof-term and kernel rewards, process and decomposition rewards, counterexample and abstention rewards, calibration, cost, novelty, vector rewards, Pareto selection, and defenses against reward hacking. Reward never substitutes for proof or authorization.

Chapters 35 through 46 address curriculum graphs, adversarial task generation, self-play, proof-counterexample games, debate as candidate generation, learning progress, human-guided curricula, tool synthesis, adapter repair, architecture search, schema evolution, benchmark co-evolution, and sandboxed candidate evaluation.

Chapters 47 through 58 treat continual learning, memory consolidation, replay, theorem and skill distillation, meta-learning across worlds, transfer, specialization, catastrophic forgetting, regression control, causal graphs, off-policy evaluation, module influence, interaction effects, cross-module credit, delayed reward, and nonidentification.

Chapters 59 through 70 apply descent to evolution itself. Contexts form covers; local policies act as sections; overlaps require compatibility treaties; cocycle and higher-coherence tests expose obstructions; covers may be refined; compatible policies may be glued only under an effectivity procedure; version bundles are dependency-aware; whole-system replay precedes external promotion requests; shadow and canary stages remain distinct from signed atomic promotion and rollback; root upgrades require successor epochs.

The final ten chapters cover adversarial security, human mathematical review boards, incidents, retraction, correction, distributed training, module-local and whole-system benchmarks, compute and energy accounting, reproducibility, contamination control, end-to-end recursive discovery campaigns, scientific delivery, and the long-horizon roadmap.

Volume VI does not relax the constitution of Volume V. An evolving module can generate a candidate, assemble evidence, abstain, or request evaluation. It cannot issue its own proof receipt, sign its own promotion, select its own authoritative predecessor, or declare its own recovery complete.

From foundations to architecture, algorithms, modules, and evolution

The six-volume progression contains six explicit stages:

$$\begin{aligned}&\text{source-critical mathematical foundation}\\ &\longrightarrow \text{typed world architecture}\\ &\longrightarrow \text{classified algorithms}\\ &\longrightarrow \text{modular implementation}\\ &\longrightarrow \text{reproducible publication and release}\\ &\longrightarrow \text{governed evolution.}\end{aligned}$$

Stage one, represented by Volumes I and II, establishes source-critical mathematical foundations. Stage two, Volume III, turns worlds, bridges, claims, assumptions, provenance, and authority boundaries into typed architecture. Stage three, Volume IV, classifies executable procedures as exact, certificate-producing, bounded, heuristic, or research. Stage four, Volume V, assigns those procedures to modular services and immutable artifacts. Stage five, also completed through Volume V, governs reproducible publication, independent review, release, and correction. Stage six, Volume VI, develops learning and co-evolution under the inherited authority structure.

The arrows express typed inheritance. Mathematical concepts retain meanings richer than their schemas; schemas prepare algorithms; algorithms receive implementations; implementations produce evidence; publication renders admitted records; evolution improves candidate-producing processes.

Consider descent. Volume I studies local data, covers, overlaps, cocycles, and effectivity in specified categories and topologies. Volume II follows descent through higher, perfectoid,

v-sheaf, prismatic, and condensed settings. Volume III adopts a task-organization pattern of local sections and overlap obligations. Volume IV supplies finite sheaf and descent certifiers. Volume V supplies local artifacts and compatibility checkers. Volume VI uses local-policy and version-bundle language.

The software, publication, and evolution uses are **bounded structural analogies** whenever they lack a declared site, cover family, restriction maps, overlap objects, cocycle conditions, and an effectivity theorem or procedure. With those data present, a chapter may state a precise mathematical or project-native descent structure. With decomposition and recombination alone, the accurate terms are locality, compatibility, and assembly. This convention preserves the mathematical force of *descent* while retaining its value as a design pattern.

The same progression applies to representability. In the foundational volumes, representability creates geometric objects from functors under theorem-specific hypotheses. The architecture records a representability gate; algorithms test supported finite or syntactic fragments; modules manage the evidence; publication preserves the exact scope; evolution improves strategy selection while existence remains governed by the mathematical criterion.

For six-operation formalisms, the foundation supplies functors, adjunctions, exchange maps, projection formulas, and dualities. The architecture records a coherence cube. Algorithms generate typed obligations or evaluate supported fragments. Modules dispatch those obligations to exact backends. Publication preserves the theorem scope, and evolution improves routing or lemma selection. Universal evaluation requires an independently established algorithmic domain.

For motives, the foundational volumes distinguish conjectural categories, established constructions, triangulated and homotopical approaches, realization functors, and open comparison faces. MUE organizes structured finite Nori-style proxies. Algorithms manipulate finite diagrams, endomorphism algebras, idempotents, and realizations under explicit bounds. Modules and releases preserve the qualifier *finite proxy*. Evolution may propose alignments or bridge hypotheses with status `R`.

For anabelian reconstruction, theorem-scoped established results occupy one plane; project-native theater and comparison research occupy another; literal-IUT definitions and disputes occupy a third. Architecture, algorithms, modules, publication, and evolution preserve these namespaces. Established arithmetic closure requires independently admitted theorem evidence for every disputed edge.

This layered translation yields a general design principle: **semantic inheritance grows in explicitness while authority follows independent evidence**. Downstream representations expose assumptions, bounds, provenance, relation strength, and failure modes. A formal target may be highly precise and still require fidelity review. A deterministic program may reproduce an irrelevant quantity exactly. A signed release authenticates bytes while mathematical validity depends on the recorded proof route. Precision of representation and strength of warrant are orthogonal.

The architecture separates six practical roles:

1. mathematical sources define or support claims;
2. compilers and generators produce representations and candidates;
3. exact or bounded engines compute and check registered predicates;
4. independent evaluators assemble scoped evidence;
5. designated authorities decide status, publication, activation, and rollback transitions;
6. human mathematicians and editors exercise final scholarly judgment.

Agreement among candidate-producing roles remains evidence for the designated decision route. This is the operational consequence of the no-self-certification principle and the intellectual consequence of source-critical mathematics.

Cross-volume inheritance is recorded through an explicit artifact chain:

$$\begin{aligned} &\mathsf{SourceClaim} \longrightarrow \mathsf{NormalizedClaim} \\ &\longrightarrow \mathsf{WorldScopedTask} \\ &\longrightarrow \mathsf{CandidateArtifact} \longrightarrow \mathsf{IndependentReceipt} \\ &\longrightarrow \mathsf{StatusDecision} \longrightarrow \mathsf{PublicationArtifact}. \end{aligned}$$

Each arrow changes the kind of object under consideration. A `SourceClaim` preserves wording, locator, authorship, edition, and context. A `NormalizedClaim` exposes binders, hypotheses, conclusion, variance, universes, and relation strength while retaining source spans. A `WorldScopedTask` declares the ambient theory, supported fragment, requested operation, and allowed premises. A `CandidateArtifact` may be a proof term, computation, counterexample, diagram, bridge, formal declaration, or failure report. An `IndependentReceipt` records the predicate checked by a named verifier. A `StatusDecision` is an authority event over an exact subject and evidence closure. A `PublicationArtifact` renders that record under a release identity.

The intermediate identities make characteristic errors observable: target drift between source and proof search, absent replay between candidate and status, missing attribution review before publication, or publication appearance used as epistemic evidence. Counterexamples, corrections, and source revisions create branches while preserving the chain.

The chain also shows where human judgment enters. Human mathematicians and editors select source interpretations, assess normalization fidelity, evaluate significance, adjudicate disputed evidence, and authorize publication. They address open-textured scholarly acts: choosing the intended theorem among source variants, evaluating historical responsibility, assessing the naturality of assumptions, and shaping exposition at the warranted strength.

Mechanical strictness complements that judgment. Digests establish byte identity. Parsers detect omitted binders in supported grammars. Kernels replay proof terms. Exact solvers verify finite certificates. Dependency graphs propagate staleness. Build systems compare ordered manifests.

The architecture combines exact automation with explicit scholarly interpretation. Replayable checks cover formal predicates; sourced interpretive acts cover historical and semantic judgments; `O` records missing evidence; `Q` preserves genuine disputes; measured learning improves candidate production under externally governed standards.

A six-stage theorem pathway

A theorem pathway makes the six stages concrete. Suppose a chapter needs a proper-base-change statement for a specified coefficient system. At stage one, the mathematical foundation identifies the geometric world, morphism class, coefficients, finiteness assumptions, derived category, comparison map, theorem source, expose author where relevant, and exact conclusion. Source criticism also distinguishes the classical theorem from later generalizations.

At stage two, the typed architecture creates a claim record and world descriptor. The source and target of the exchange map, variance of every functor, universe bounds, dependency closure, and status become machine-readable fields. The record includes a preservation profile and any open compatibility cells. The theorem's familiar name serves navigation; the typed statement serves execution.

At stage three, an algorithm may compile the claim into a formal task, check source-to-target structure, generate Beck-Chevalley obligations, or evaluate a finite instance. Each algorithm announces its class. A certificate-producing compiler can establish a faithful parse within its grammar. A bounded checker can establish a finite diagram. A theorem prover can produce a candidate proof term. Their outputs converge only through the appropriate independent receipts.

At stage four, modular services resolve immutable source spans, elaborate the formal declaration, dispatch proof or computation tasks, preserve provenance, and store failures. A proof adapter binds the proof object to a named kernel and environment. A target-fidelity component compares the source claim with the formal target. A dependency service records the axiom and theorem closure. Capability policy keeps candidate generation, verification, signing, and promotion in separate roles.

At stage five, publication assembles the admitted statement, proof status, source locator, notation, figure references, and obligations from one ordered source. PDF and DOCX render the same semantic claim. Build receipts establish source and environment identity. Mathematical review, source review, visual review, and release authorization each contribute a distinct disposition.

At stage six, evolution may improve retrieval of the theorem, selection of a formal library lemma, decomposition of the proof task, or routing to a stronger checker. It may propose a more efficient compiler pass or a clearer exposition. The theorem's source identity, formal target, kernel policy, and promotion authority remain fixed within the governing epoch. A proposed change enters as a successor candidate with replay and regression evidence.

The pathway also supports correction. Discovery of a source mismatch creates an `O` record, suspends affected reuse, and triggers a corrected normalization. Discovery of a genuine published dispute creates a `Q` record for the disputed assertion and preserves the competing evidence. A kernel regression marks receipts stale under the affected environment. A publication correction creates a successor release. Each event propagates through the exact dependency closure.

The six-stage model therefore connects mathematical scholarship and implementation through typed transformations. It gives every participant a clear question: What exact proposition is being asserted? Which world gives it meaning? Which procedure acted on it? Which module produced the artifact? Which review admitted it? Which components may learn from the event? The answers form a durable lineage from source to evolving system.

Epistemic constitution and claim-status discipline

The monograph uses seven atomic epistemic statuses:

Code	Meaning
`E`	established theorem, definition, or bibliographic fact within an exact checked scope
`S`	source-grounded synthesis, interpretation, or structural organization
`D`	project-native definition, schema, interface, or constitutional rule
`R`	falsifiable research construction, bounded program, or experimental capability
`C`	precisely stated conjecture

Code	Meaning
O	unresolved source, proof, coherence, implementation, security, review, or integration obligation
Q	disputed, unsettled, or quarantined claim

These symbols are disjoint constructors rather than grades on one scale. D identifies a definition at every implementation stage. C identifies a conjecture. O records missing source, proof, review, implementation, or integration evidence. Q preserves a genuine dispute, unsettled verdict, or quarantine decision while leaving truth open. S identifies scholarly synthesis at its proper evidentiary level.

Statuses attach to exact assertion aspects. A chapter may introduce a schema as D, propose a capability as R, and record missing implementation evidence as O. A compact display such as D/R/O presents several assertions. The same sentence in different sources may carry different statuses because its hypotheses, intended world, interpretation, or authority differ.

Epistemic status is orthogonal to workflow state. An artifact can be draft, candidate, parsed, compiled, computed, checked, proved, refuted, published, withdrawn, superseded, or stale. A formally checked transcription of a conjecture remains C. A research algorithm can produce an exact result for one finite input while the general capability remains R. A theorem can become stale as a software artifact after a dependency update without becoming mathematically refuted. Publication is a workflow event, not an epistemic constructor.

An established assertion requires an exact subject. The record preserves the proposition, hypotheses, quantifiers, ambient world, universe discipline where relevant, relation strength, scope, source identity, edition or version, locator, and dependency closure. Theorem names provide navigation; expose and theorem locators provide evidence. Modern generalizations receive separate claim identities from classical statements.

Formal proof introduces further distinctions. A source statement is compiled into a formal target. Fidelity review compares binder order, hypotheses, quantifiers, conclusion, variance, universe constraints, and allowed premises. A named, pinned kernel then replays a proof candidate in a recorded environment. The complete axiom closure remains visible, and the proof receipt binds the target hash and environment. Publication and attribution proceed through their own decisions. Kernel replay proves the formal target under the recorded assumptions; a target-fidelity receipt connects that target to the intended source.

Computation has its own ceiling. An exact computer-algebra result, finite-model search, sheaf check on a finite site, or exhaustive bounded search can establish the registered finite predicate. A separate proof can connect that predicate to a general theorem. Timeout records resource exhaustion. An exhausted counterexample search records its searched domain. A found counterexample refutes a universal statement when every hypothesis and the input interpretation are verified.

Statistical systems occupy a proposal role. Retrieval can identify candidate sources. A language model can propose a formalization, proof plan, lemma, counterexample, notation, or new world. Multi-agent debate can diversify candidates. Scores can rank them. None of these operations changes truth status. Confidence, consensus, novelty, citation frequency, benchmark performance, and rhetorical fluency have no theorem authority.

The dependency system is role-sensitive. E claims support established conclusions within their exact scope. S supports exposition and design lineage. D supports formation and typing. R and C support explicitly conditional research branches. O identifies a required witness

before promotion. `Q` supports dispute analysis and quarantined conditional study while remaining outside established closure. Dependency admission evaluates each role directly.

Relation strength remains separate from status. Equality, isomorphism, equivalence, homotopy, weak equivalence, comparison map, reconstruction up to inner automorphism, and agreement through a truncation are different claims. An `E` equivalence retains the relation *equivalence*. A `D` field named `is_equivalent` is a schema element whose witness arrives separately. A checked square establishes its declared finite cell. Higher coherence and effectivity require their stated data.

The constitution preserves history. Corrections create successor records. Retractions withdraw reliance within a declared scope while retaining the earlier event. Supersession records replacement and leaves causal interpretation to evidence. Rollback changes an active system assignment; publication correction follows its own route. An incident record preserves the original claim, evidence, failure, containment, review, correction, and recovery events.

The no-self-certification principle assigns candidate production, proof checking, signing, promotion, and rollback to distinct authority roles. Repeated execution by one model remains same-principal evaluation. Independence may require distinct executable versions, mutable state, data access, credentials, execution domains, and institutional roles appropriate to the claim. Candidate and evaluator agreement contributes evidence to the designated authority.

The purpose of this discipline is semantic precision. It distinguishes:

- source familiarity for exact citation;
- type correctness for semantic fidelity;
- compilation for proof;
- proof search for proof receipt;
- computation for theorem;
- repeated success for generality;
- confidence for truth;
- publication for verification;
- signature for mathematical validity;
- rollback for retraction;
- consensus for authority;
- richer language for solved obstruction.

The monograph asks the reader to treat these distinctions as part of the mathematics and as durable publication data.

Reading routes for different communities

The integrated order is canonical, while readers may enter through different thematic routes. These routes preserve dependency direction: later contracts inherit meanings established earlier.

For mathematicians

A mathematician interested primarily in algebraic geometry, topology, arithmetic geometry, or category theory should begin with the Volume I prospectus and the source-critical opening chapters, then enter the relevant thematic part. Scheme theorists may proceed through relative geometry, descent, and six operations. Homotopy and higher-category readers may move from topoi and stacks to *Pursuing Stacks* and then to Volume II's higher, derived, and spectral parts. Arithmetic geometers may follow etale cohomology and anabelian reconstruction into p-adic comparison, perfectoid, prismatic, and arithmetic-moduli chapters.

After reading a mathematical corridor, the reader should consult the corresponding synthesis instruments near the ends of Volumes I and II. These reveal how the monograph encodes hypotheses, transformations, and missing comparison faces. Volume III is then best read selectively: mathematical-world specifications, typed maps, epistemic status, the operational rising sea, proof receipts, and the architecture associated with the chosen topic. Volume IV supplies the boundary between the mathematical theory and its effective fragments. This route makes clear where the monograph is proving or reporting established mathematics and where it is proposing a computational research program.

Readers concerned with historical attribution should follow source chapters before thematic chapters. The source layer explains why EGA, SGA, FGA, manuscript programs, modern formulations, and retrospective accounts require different citation practices.

For AI researchers

AI researchers should begin with the integrated preface, then read Volume III Chapters 1 through 13 on machine obligations, world representations, status typing, artifact envelopes, scope, proof receipts, and dependency closure. They should continue through the GMUS and GMLS contracts, paying special attention to the separation among world creation, reasoning within a world, language formation, retrieval, formal checking, and publication.

Volume IV's first two parts are essential. They define effective fragments, algorithm classes, abstention, failure artifacts, source-to-IR compilation, target fidelity, and independent replay. The later algorithm parts can then be sampled according to interest: categorical computation, proof search, counterexample generation, multi-agent candidate production, or benchmark constitution.

Volume V's constitutional chapters prepare the reading of Volume VI. Reward design, hierarchical policies, tool routing, curricula, debate, and co-evolution operate under fixed proof, signing, promotion, and rollback authorities. Reinforcement-learning readers should focus on the distinctions between reward and truth, evaluation and certification, and local compatibility and global effectivity.

The mathematical volumes supply essential meanings for *descent*, *equivalence*, *realization*, and *reconstruction*. AI work should carry those theorem-level meanings into its representations.

For formalizers and proof engineers

Formalizers should begin with the source and theorem-lineage chapters of Volume I, then read the notation and status conventions in this preface. Volume III's artifact, world, formal-task, target-fidelity, proof-receipt, and dependency contracts form the principal architectural route. Volume IV's source-to-IR compiler, target-fidelity guard, proof replay, axiom closure, universe checking, diagram obligations, and finite categorical algorithms provide the executable route. Volume V's Lean, Coq, Isabelle, SMT, model-finder, CAS, and verified-numerics adapters show how multiple backends can coexist without merging trust.

Particular care should be given to relation strength and target identity. Alpha-renaming and approved normalization may preserve a target; hypothesis deletion, implication reversal, theorem weakening, undeclared axioms, and conversion of equivalence into equality create different targets. A successful kernel replay is meaningful when bound to the approved formal target and exact environment.

Formalizers may also use the volume maps to identify realistic fragments. Finite sites, finite categories, bounded complexes, explicit rings, exact algebra, and theorem-scoped

adapters offer tractable targets. Full motives, unrestricted higher coherence, general six-functor evaluation, and literal formalization of disputed theories remain research boundaries.

For software and systems engineers

Engineers should begin with Volume V Chapters 1 through 12, which define immutable roots, content-addressed artifacts, authority separation, schemas, events, and the inert evolution interface. They should then read the storage, workflow, API, security, observability, deployment, and publication chapters. Volume III supplies the normative reasons for these components; Volume IV supplies the algorithms and their exact scope.

The key engineering principle is that data integrity, mathematical validity, and authorization are distinct predicates. Schema validation establishes conformance. A signature establishes policy-scoped authenticity. A reproducible build establishes a replay relation. Mathematical truth and source completeness follow their dedicated checks. Evaluator reports feed a separately authorized status transition.

Engineers implementing experimental slices should prefer exact small kernels and explicit failure unions. The finite-poset-site sheaf and descent certifier has finite inputs, explicit matching families, existence and uniqueness checks, counterexample outputs, and proof obligations. Its claim scope is the registered finite fragment.

For editors, historians, and publication specialists

Editors should begin with the source-critical genealogy in Volumes I and II, the epistemic constitution, and Volume V's publication chapters. Every attributed claim should be checked against the smallest supporting source unit. Expose authors remain visible beside seminar-volume direction. Each transcription or edition keeps its own pagination. Modern generalizations receive separate records from classical statements. Disputed verdicts are attributed as positions.

Publication specialists should treat the integrated manifest as the authoritative order. PDF and DOCX are two renderings of one semantic publication. Volume-level outputs are proofing views of that source. Formula inventories, figure identities, captions, alt text, theorem labels, cross-references, metadata, hashes, and source order require parity even when pagination differs.

Editors responsible for prose should preserve status language. Words such as *proves*, *implements*, *validates*, *certifies*, *reconstructs*, and *equivalent* have contractual force. A change from *proposes* to *establishes* changes the claim. Qualifiers such as *finite*, *bounded*, *truncated*, *theorem-scoped*, and *under the stated hypotheses* preserve scope.

For interdisciplinary reading groups

An interdisciplinary group may organize a repeated five-step cycle:

1. read one mathematical chapter from Volumes I or II;
2. identify its world change, preservation claims, and source lineage;
3. read the corresponding architectural contract in Volume III;
4. inspect the algorithmic fragment in Volume IV and module assignment in Volume V;
5. identify in Volume VI the components that may evolve while truth conditions remain fixed.

This route exposes the central thesis of the monograph: mathematical intelligence advances through constructing, comparing, and governing worlds in which answers have exact meaning.

Notation, source, and proof conventions

The six volumes span mathematical traditions with different notational habits. A layered convention supplies uniformity while preserving source distinctions.

Grothendieck universes are written $\mathbb{U} \in \mathbb{V}$ when size discipline matters. Objects declared $\mathbb{U}$-small may form categories or functor categories in $\mathbb{V}$. This convention is an editorial and architectural interface. Historical sources retain their own foundational formulations.

Categories are written $\mathcal{C}, \mathcal{D}, \ldots$; functors as $F : \mathcal{C} \to \mathcal{D}$; natural transformations as named arrows when their direction matters. Opposite categories, variance, and base categories are stated explicitly. A Cartesian square records a candidate or specified pullback situation; property preservation follows the relevant theorem.

Equality $=$, isomorphism $\cong$, equivalence $\simeq$, weak equivalence, homotopy, and higher invertible cells are distinct relation types. Normalized theorem records state the intended relation explicitly. The same rule applies to implication, equivalence of propositions, reconstruction up to inner automorphism, and identity after truncation.

Sites, topologies, and covers are named whenever local-to-global claims depend on them. Cech notation identifies a complex or nerve; a comparison theorem connects it to derived cohomology. Pairwise compatibility, cocycle coherence, and effective descent are successive conditions. Hyperdescent and hypercompletion are stated when required.

Derived and higher objects retain their enhancement. Passing to a homotopy category, cohomology group, truncation, or classical shadow is a declared operation. A statement proved after truncation remains scoped to that truncation. Spectral sequences carry page, differential, filtration, and convergence conditions. Complexes carry boundedness and coefficient conventions where they matter.

Realization and comparison arrows are directed. A comparison isomorphism under a period ring or completion has that exact scope. A realization prism may have open faces. Commutativity and compatibility require their own sources or obligations.

Machine-facing names such as GMUS, MUE, TRE, TNCA, TNRA, GRA, GMLS, GCA, GDRA, DDA, and AURORA are project-native definitions. Their names signal mathematical lineage; authorship of the software architecture belongs to this monograph.

Source references follow a hierarchy. Primary sources govern origin and attribution. Secondary sources support exposition, historical interpretation, later proof, correction, and modern formulation. A complete source identity may include author, title, publication or manuscript form, edition, volume, expose, expose author, section, theorem, proposition, page, date, stable URL or DOI, and a local immutable digest. Unavailable fields become $\circ$.

The work distinguishes a **source family** from a **claim locator**. *EGA*, *SGA*, *FGA*, *Pursuing Stacks*, and *the Stacks Project* identify families; an established theorem dependency uses the narrowest stable locator available. Revised sources carry version identity. Stronger modern theorems receive separate records.

Quotations are used sparingly. Translation is marked as translation, and edition-specific wording stays attached to its witness. This preface paraphrases methodological themes. Italics signal titles or emphasis; explicit quotation marks and locators signal historical quotation.

Proof presentation has four principal forms.

1. A complete proof may be given in the prose, with hypotheses and dependencies visible.
2. A theorem may be cited to an exact source, with enough statement detail to prevent scope drift.
3. A formal proof may be represented by a target-fidelity record and an independently replayed kernel receipt.

4. A computational claim may be supported by a bounded certificate, exhaustive witness, or reproducible exact computation.

Sketches, heuristics, diagrams, and examples are identified as such. A complete proof supplies every required step or exact dependency. Computer output used as proof evidence records the checked predicate, environment, certificate, and replay route. An external theorem prover's success is interpreted within the supplied target and axiom closure.

Definitions introduced by the monograph are typographically and epistemically separated from established mathematical definitions. A *Mathematical World Descriptor*, *Descent Ledger*, *Realization Prism*, *Algorithm Contract*, *Artifact Envelope*, or *Module Evolution Context* is a project-native instrument. It may organize established material faithfully; mathematical warrant continues to follow the stored sources and proofs.

Figures are explanatory artifacts with their own identities, captions, alt text, sources, and lineage. A caption states whether an arrow encodes conceptual dependence, a proposed flow, a theorem implication, or a software message. Status is written in text as well as color. Mathematical commutativity follows its proof or receipt. Reformatting preserves the identity of the underlying evidence.

Limits, disputed claims, and research boundaries

The scope of the monograph makes explicit limitation indispensable. Six boundaries organize the claim surface.

First, mathematical coverage is selective despite its scale. The volumes provide a structured map and selected theorem corridors through EGA, SGA, motives, infinity-categories, p-adic Hodge theory, condensed mathematics, and anabelian geometry. Complete hypotheses and proofs remain in the exact primary and modern sources.

Second, structural analogy has a bounded role. Software-task gluing, publication assembly, and policy co-evolution are described as descent analogies until site, cover, restriction, overlap, cocycle, and effectivity data are supplied. Topos, motivic, and anabelian language enters system architecture through project-native definitions whose mathematical strength follows any accompanying functor and theorem.

Third, computational scope is fragmentary. General theorem proving includes undecidable regions; many categorical constructions lack effective presentations; higher coherence can exceed feasible bounds; CAS and SMT procedures have theory-specific domains; finite-model searches cover declared finite spaces; numerical evidence supports numerical claims. Volume IV therefore classifies procedures as exact, certificate-producing, bounded, heuristic, or research. The class forms part of every result.

Fourth, specification and implementation have separate evidence. Volumes III through VI contain normative architectures, proposed contracts, traceability hooks, and reference-platform designs. Production status follows allocated components, source paths, executables, tests, security review, replay receipts, and release evidence.

Fifth, self-evolution remains governed and incomplete. Volume VI assigns certification, root changes, promotion, and rollback to external authority roles. Causal attribution across interacting modules may remain unidentified; rewards admit gaming risks; evaluation data admit contamination risks; long-horizon outcomes complicate credit assignment; overlap compatibility may leave global effectivity open. These are central research boundaries.

Sixth, disputed mathematics remains disputed. The IUT treatment records established bibliographic facts, literal source definitions, claimed implications, published critiques, responses, and dated evidence assessments. Correctness follows proof-edge review rather than

venue, reputation, consensus, or rhetorical force. Claimed arithmetic consequences remain quarantined while required proof edges await independent admission.

Standard conjectures, mixed motives, section conjectures, open reconstruction problems, unrestricted motivic six operations, some higher coherence programs, and proposed machine calculi retain their open status. Chapter and interface names provide research addresses; mathematical resolution follows evidence.

The monograph uses restrained claims of significance. Grothendieckian methods have transformed major regions of mathematics. Formal verification greatly increases assurance for a precisely stated target while specifications, source interpretation, kernels, libraries, and hardware retain review obligations. Artificial intelligence can accelerate search and synthesis; warrant follows the evidence chain. The architecture is offered for criticism, implementation, formalization, and experiment.

Research boundaries are represented positively through obligations. An unsupported higher cell becomes a coherence obligation. A missing comparison becomes an open face. A disputed implication becomes a quarantined edge. A finite approximation records its bound. Source uncertainty becomes an attribution obligation. A security assumption becomes a test. A causal nonidentification result constrains experimental claims. Incompleteness thereby becomes structured information.

Publication and reproducibility method

The publication method follows the same principles as the mathematical architecture. One exact ordered manifest resolves the common front matter, six volume dividers, six volume manifests, and common back matter. PDF and DOCX render the same semantic source closure. Pagination may differ while source membership, order, formulas, theorem labels, figures, captions, citations, volume boundaries, and authorial identity remain aligned.

The canonical identity is fixed: *GAI-MUA: The Grothendieckian Machine*, authored by Vincent Liu, under the WiseMotif.ai institutional and imprint identity, in the *WiseMotif Monograph of Mathematical Intelligence* series. The six volumes form one work. Per-volume artifacts are proofing and distribution views of the integrated release.

The Markdown source is designed for Pandoc-based PDF and DOCX production. It favors semantic headings, native mathematics, tables, figures, captions, cross-references, and restrained structural markup. Editorially justified filters handle format-specific pagination. DOCX remains editable, unencrypted, and structurally live; PDF provides fixed-layout reading and print evidence. Both formats carry live semantic content appropriate to their medium.

Publication is an evidence-bearing transformation with a strict ceiling. A successful build establishes that a named toolchain produced artifacts from named inputs under a named environment. Mathematical proof, source review, figure review, and release authorization retain their dedicated evidence routes.

Reproducibility is exact-subject reproducibility. A build or experiment manifest records ordered source identity, dependency closure, tool versions, filters, styles, fonts, figures, environments, bounds, outputs, tests, failures, and hashes. Mutable labels such as *latest* or a branch name serve discovery; immutable revisions and enumerated local inputs support replay.

An independent replay compares outputs under a declared rule. Byte identity may be required for deterministic artifacts; semantic or structural comparison may be appropriate where container metadata or rendering engines introduce controlled variance. The comparison rule is part of the claim. Hash comparison and accessibility review answer different predicates.

Reproducibility and validity are separate coordinates. A theorem, benchmark, compiler configuration, or source manifest may replay exactly while its scientific evaluation raises other findings. Contamination control records whether protected evaluation information crossed a declared data, custody, feedback, model-lineage, or purpose boundary. An audited graph supports the bounded statement `no_known_exposure_within_audited_closure`.

The release method uses successor identities. Staging, independent review, final build, post-build inspection, and atomic promotion are distinct events bound to exact manifests and hashes. Builder, reviewer, and admitting authority remain distinct. Corrections create successor releases and preserve predecessor evidence.

The publication design aims at the restrained legibility associated with the strongest research-monograph tradition while retaining an original WiseMotif.ai visual system. Typography, theorem presentation, indexes, notation apparatus, source concordances, figure lineage, accessibility, and PDF/DOCX parity support the transmission of mathematical structure across media and preserve trade-dress independence.

Concluding synthesis

The six volumes advance one governing proposition: a serious architecture for mathematical intelligence represents statements together with the worlds in which they are meaningful and the transformations by which those worlds are related.

From Grothendieckian mathematics the work inherits relational definition, universal construction, relative viewpoint, locality, descent, categorical height, comparison, obstruction, and reconstruction. Source criticism contributes exact source, version, hypothesis, and authorial-unit discipline. Formal methods contribute target identity, kernel replay, axiom visibility, and failure certificates. Systems engineering contributes immutable artifacts, capability separation, provenance, observability, reproducibility, and rollback. Machine learning contributes candidate generation, adaptive routing, curricula, and policy improvement under a separate truth-authority layer.

The resulting machine is intentionally federated. Schemes, motives, proof receipts, historical sources, disputed claims, and deployment states retain distinct types. Every bridge has direction and loss. Every compressed object retains a route to provenance. Learning operates under the constitution that judges it. Automation includes unknown and abstention as legitimate results.

The deepest mathematical continuity across the work is descent: local data yield global objects under compatibility and effectivity. Architecture, publication, and evolution reuse that pattern as a bounded structural analogy until they supply corresponding site, cover, restriction, cocycle, and effectivity data. In every setting, overlap and return obligations carry the rigor.

The corresponding deepest discipline is typed distinction. Source statement, interpretation, formal target, proof candidate, kernel receipt, publication, and social acceptance are related through explicit edges. Object, shadow, invariant, realization, and reconstruction retain their relation strengths. Candidate, evaluator, signer, promoter, and rollback authority retain their roles. Schemas and ledgers make productive relations possible while preserving the distinctions on which mathematical truth depends.

The work is offered as an auditable collaboration among mathematical imagination, historical scholarship, formal proof, exact computation, software engineering, and adaptive machine assistance. Its success criterion is the preservation of conditions under which certainty, uncertainty, disagreement, correction, and discovery can each be stated truthfully.

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This monograph is written in gratitude to Alexander Grothendieck for the mathematical ideas, methods, programs, and philosophical reflections that made the construction of new mathematical worlds a central and explicit mode of modern thought. The work owes particular intellectual debts to the relative viewpoint, schemes, sites and topoi, descent, cohomological architectures, moduli and representability, motives, anabelian programs, higher-categorical aspirations, and the reflective vocabulary through which Grothendieck described mathematical creation.

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Exact local attribution remains in the source concordances and chapter-level records. The present acknowledgement expresses intellectual gratitude and preserves the distinction between local authorship, redaction, correspondence, editing, criticism, and later theorem formation.

No person or institution named here has reviewed, authorized, sponsored, or endorsed this work unless expressly documented.

The acknowledgement implies no personal collaboration with Grothendieck or with the mathematicians named here. The project-native expressions *Grothendieckian Machine*, *Diagrammatic Poetry Thinking*, GMUS, GMLS, and the associated computational architectures are the author's constructions, offered as interpretations and research instruments inspired by mathematical ideas and methods in the cited lineages.

Gratitude also extends to the documented preservation, editing, translation, digitization, teaching, criticism, and formalization work cited throughout the source dossiers. These roles make source-critical synthesis possible and supply standards for judging the monograph's claims.

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Chapter 1. The architecture and publication history of EGA

Structural and mathematical principles

Structural Principle

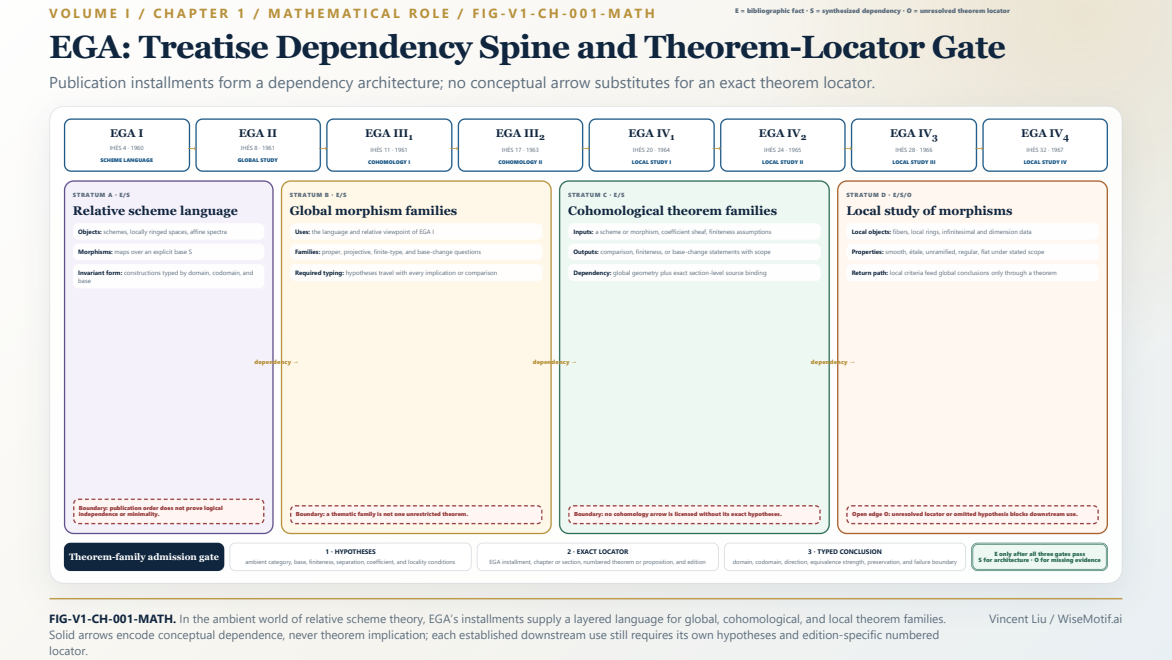

Figure 1: Volume I, Chapter 1 - Structural principle

FIG-V1-CH-001-MATH. In the ambient world of relative scheme theory, EGA's installments supply a layered language for global, cohomological, and local theorem families. Solid arrows encode conceptual dependence, never theorem implication; each established downstream use still requires its own hypotheses and edition-specific numbered locator.

Vector PDF · Provenance: `FIG-V1-CH-001-MATH-THE-ARCHITECTURE-AND-PUBLICATION-HISTORY-OF-EGA-MATH`

Mathematical Principle

FIG-V1-CH-001-MATHEMATICAL-PRINCIPLE-R1

Vector PDF · Provenance: `FIG-V1-CH-001-CORE`

Author: Vincent Liu

Institution and imprint: WiseMotif.ai

Chapter ID: `V1-CH-001`

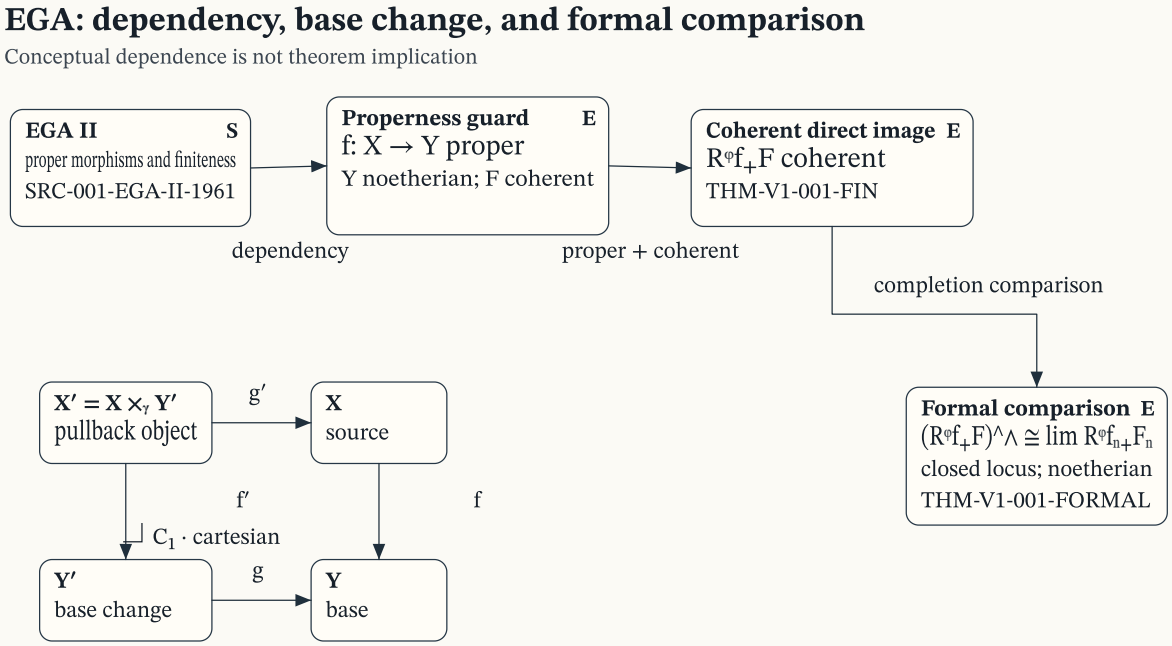

Figure 2: Volume I, Chapter 1 - Mathematical principle

1.0 Contract, epistemic grammar, and boundary

This chapter has one deliberately bounded task: to explain how the published *Éléments de géométrie algébrique* formed a mathematical architecture and how that architecture was distributed across its publication history. It is not a biography of Alexander Grothendieck or Jean Dieudonné, not a survey of the Séminaire de géométrie algébrique, and not a compressed replacement for the later chapters of this volume. Breadth of coverage does not license repetition, decorative history, or theorem statements without hypotheses.

The chapter uses atomic epistemic records:

- `E` records an established textual fact, definition, theorem, or elementary consequence whose exact source is supplied.
- `S` records a source-grounded interpretation or comparison.
- `D` records a chapter-local definition, data type, or interface.
- `R` records a bounded research, audit, or formalization program.
- `O` records an independently dischargeable obligation.
- `C` records a conjectural mathematical claim.
- `Q` records a disputed, incomplete, edition-sensitive, or interpretive claim.

Each handle has exactly one of these statuses. An interpretation does not become a theorem because it is plausible; a bibliographic fact does not prove a mathematical implication; and a source-addressing scheme does not prove the claim stored at that address.

Status `E-001-SCOPE`; claim `CLM-V1-001-SCOPE`. The original journal sequence published under the EGA title consists of EGA I, EGA II, EGA III in two parts, and EGA IV in four parts, appearing in *Publications mathématiques de l'IHÉS* between 1960 and 1967. The title pages identify A. Grothendieck and use the formula "Rédigés avec la collaboration de J. Dieudonné." The precise issue, year, and page records appear in §1.2.

Status `S-001-THESIS`; claim `CLM-V1-001-THESIS`. The most useful way to read the published EGA is neither as eight unrelated fascicles nor as a linear encyclopedia. It is a dependency architecture with four interacting strata: language and relative objects; global

classes of morphisms and constructions; coherent cohomology and its variation; and local properties of schemes and morphisms. This four-stratum description is an interpretation supplied by this chapter. It is not an EGA theorem and is not claimed to be the only valid table of contents.

Status `Q-001-AUTHORSHIP`; claim `CLM-V1-001-AUTHORSHIP`. Modern bibliographic systems do not encode the collaboration formula uniformly. Some database records list Grothendieck as the sole author, while conventional mathematical citations often name Grothendieck and Dieudonné together. This chapter therefore preserves the title-page wording and records database authorship separately; it does not infer a private division of intellectual or editorial labor from either convention.

1.0.1 Non-preemption

Chapter 2 owns the seminar ecology and thematic range of SGA. Chapter 3 owns FGA and the functorial creation of moduli. Chapters 7 and 8 own source-version control and machine-readable theorem lineage. Chapters 13–30 own the detailed mathematics of schemes and relative geometry, and Chapters 31 onward own sites, topoi, descent, cohomology, stacks, motives, and reconstruction. The present chapter may identify interfaces to those subjects, but it does not provide their full definitions, theorem spines, or proofs.

In particular, an EGA cross-reference to a projected Chapter V or VI is historical evidence about the planned treatise. It is not evidence that the projected chapter was published under the EGA title. Material later developed in FGA, SGA, or other works remains owned by those source families. Similarity of subject matter does not authorize silent substitution.

1.0.2 Universes and mathematical ambient world

Status `D-001-UNIVERSE`; definition `DEF-V1-001-UNIVERSE`. For the chapter-local machine records in §§1.8–1.10, fix Grothendieck universes $\mathbb{U} \in \mathbb{V}$. Rings and affine schemes under discussion are $\mathbb{U}$-small; categories of such objects and their functor categories are formed in $\mathbb{V}$. Historical EGA statements are reported in their printed foundational language. This universe convention is an editorial interface for the present monograph, not a claim that the original fascicles used this exact notation.

For a ring A, the affine scheme is written

$$\operatorname{Spec}(A).$$

A morphism $f : X \to S$ is treated as relative data, and a change of base $g : S' \to S$ produces the Cartesian square

$$\begin{array}{ccc} X_{S'} = X \times_S S' & \longrightarrow & X \\ \downarrow & & \downarrow f \\ S' & \xrightarrow{g} & S. \end{array}$$

The square is used throughout as a typing diagram. No claim that a property is preserved by base change follows merely from drawing it; preservation requires the exact proposition defining or proving that property.

1.1 Mathematical pressure: why a treatise needed an architecture

EGA I states that the memoir and the works intended to follow it were to form a treatise on the foundations of algebraic geometry. Its introduction assumes commutative algebra, homological algebra, sheaf theory, and functorial language. It also announces a systematic

acceptance of nilpotents in local rings and a shift of emphasis from rational maps toward regular maps, or morphisms. These are textual facts, not retrospective slogans: they occur in the opening pages of the 1960 fascicle.

The pressure behind the architecture can be described without pretending that all earlier algebraic geometry had one defect. Classical varieties over a field, birational methods, coordinate calculations, and projective embeddings were powerful. They did not, by themselves, supply a single language adequate for arithmetic bases, nilpotent structure, families, infinitesimal neighborhoods, nonclosed points, and systematic change of base. The scheme language made these phenomena objects of one relative theory.

Status `E-001-NIL`; claim `CLM-V1-001-NIL`. EGA I, Introduction, journal pages 5–7, explicitly says that the systematic acceptance of nilpotent elements in local rings is among the reasons for adopting the scheme point of view, and that this move places rational mappings in a secondary position relative to regular mappings or morphisms.

Status `S-001-PRESSURE`; claim `CLM-V1-001-PRESSURE`. The architectural problem was not merely to enlarge the class of spaces. It was to coordinate four operations: formation of objects from rings, gluing of local objects, transport along a base, and expression of geometric properties in forms that can be composed, localized, or tested after base change. This interpretation is supported by the organization of EGA I and II but is not quoted from a single sentence.

The change is visible already in the affine anti-equivalence. An algebra map $\varphi : B \to A$ determines a morphism

$$\operatorname{Spec}(A) \longrightarrow \operatorname{Spec}(B),$$

so algebraic variance is reversed. EGA I, Theorem 1.7.3 and Corollary 1.7.4, identify morphisms of affine schemes with ring homomorphisms in the opposite direction. Definition 2.1.2 then defines a *prescheme* as a locally ringed space locally covered by affine schemes. The later terminological preference for "scheme" does not erase the source wording; a locator must preserve the edition's term when quoting it.

The global theory then requires more than affine gluing. A morphism must be classified by conditions that behave predictably under composition and change of base. EGA II studies affine, quasi-affine, projective, proper, finite, and related morphisms, along with homogeneous spectra, ample sheaves, valuative criteria, blowups, cones, and projective closure. EGA III adds coherent cohomology, proper direct-image finiteness, formal comparison, semicontinuity, and exchange phenomena. EGA IV turns to local properties, finite presentation, flatness, fibers, dimension, regularity, normality, limits, constructibility, smoothness, unramifiedness, and étaleness.

This sequence is architectural because each layer constrains later statements. For example, a theorem about proper direct images needs a previously typed notion of proper morphism and a coherent sheaf. A smoothness criterion needs finite-presentation and flatness language, geometric regularity of fibers, and local-ring machinery. The dependencies are mathematical, not merely chronological.

1.2 Exact source dossier and publication passport

The primary witnesses used here are the NUMDAM scans of the journal installments. Each scan begins with a machine-readable bibliographic leaf and then reproduces the title page. The journal records credit Alexander Grothendieck; the reproduced title pages read "par A.

Grothendieck" and "Rédigés avec la collaboration de J. Dieudonné." The passport therefore keeps `database_author` and `title_page_collaboration` as distinct fields.

Source handle	Installment and subtitle	Journal locator	Page span	Stable digital locator
`SRC-001-EGA-I-1960`	EGA I, *Le langage des schémas*	*Publ. Math. IHÉS* 4 (1960)	5–228	`PMIHES_1960__4__5_0`
`SRC-001-EGA-II-1961`	EGA II, *Étude globale élémentaire de quelques classes de morphismes*	*Publ. Math. IHÉS* 8 (1961)	5–222	`PMIHES_1961__8__5_0`
`SRC-001-EGA-III-1-1961`	EGA III, *Étude cohomologique des faisceaux cohérents*, première partie	*Publ. Math. IHÉS* 11 (1961)	5–167	`PMIHES_1961__11__5_0`
`SRC-001-EGA-III-2-1963`	EGA III, same title, seconde partie	*Publ. Math. IHÉS* 17 (1963)	5–91	`PMIHES_1963__17__5_0`
`SRC-001-EGA-IV-1-1964`	EGA IV, *Étude locale des schémas et des morphismes de schémas*, première partie	*Publ. Math. IHÉS* 20 (1964)	5–259	`PMIHES_1964__20__5_0`
`SRC-001-EGA-IV-2-1965`	EGA IV, same title, seconde partie	*Publ. Math. IHÉS* 24 (1965)	5–231	`PMIHES_1965__24__5_0`
`SRC-001-EGA-IV-3-1966`	EGA IV, same title, troisième partie	*Publ. Math. IHÉS* 28 (1966)	5–255	`PMIHES_1966__28__5_0`
`SRC-001-EGA-IV-4-1967`	EGA IV, same title, quatrième partie	*Publ. Math. IHÉS* 32 (1967)	5–361	`PMIHES_1967__32__5_0`

The corresponding DOI records are `10.1007/BF02684778`, `10.1007/BF02699291`, `10.1007/BF02684273`, `10.1007/BF02684890`, `10.1007/BF02684747`, `10.1007/BF02684322`, `10.1007/BF02684343`, and `10.1007/BF02732123`. A release record must bind the exact files actually consulted rather than assuming that a DOI landing page, a later reprint, and a local scan have identical pagination.

Status `E-001-PUBSEQ`; claim `CLM-V1-001-PUBSEQ`. The issue sequence is $4, 8, 11, 17, 20, 24, 28, 32$, and the publication years are $1960, 1961, 1961, 1963, 1964, 1965, 1966, 1967$. This is verified by the bibliographic leaves of the eight primary scans.

Status `E-001-PLAN13`; claim `CLM-V1-001-PLAN13`. EGA I, Introduction, journal pages 6–7, gives a provisional plan of thirteen chapters. It names Chapter I on scheme language; Chapter II on global elementary study of classes of morphisms; Chapter III on cohomology of coherent algebraic sheaves and applications; Chapter IV on local study of morphisms; Chapter V on elementary procedures for constructing schemes; Chapter VI on descent and general construction methods; Chapter VII on group schemes and principal fiber spaces; Chapter VIII on differential study of fiber spaces; Chapter IX on the fundamental group; Chapter X on residues and duality; Chapter XI on intersection theory, Chern classes,

and Riemann–Roch; Chapter XII on abelian schemes and Picard schemes; and Chapter XIII on Weil cohomology.

Status `E-001-OPENCH`; claim `CLM-V1-001-OPENCH`. The same introduction says that chapters were in principle considered open and that supplementary sections could later be issued separately. This publication model explains the repeated continuation of Chapter 0 and the splitting of Chapters III and IV into parts.

Status `S-001-INCOMPLETE`; claim `CLM-V1-001-INCOMPLETE`. The contrast between the announced thirteen-chapter plan and the published journal sequence is historically important, but "incomplete" must be typed carefully. The EGA title did not reach the announced Chapters V–XIII in the original sequence. It does not follow that their mathematical themes disappeared, that later sources simply equal the projected chapters, or that one can reconstruct an unwritten canonical continuation by merging SGA and FGA.

1.2.1 Source-passport schema

Status `D-001-PASSPORT`; definition `DEF-V1-001-PASSPORT`. An `EGASourcePassport` is the tuple

```
(source_handle,installment,part,title,
language,
database_author,title_page_collaboration,
journal,issue,
year,page_span,doi,digital_locator,
artifact_sha256,
locator_scheme,errata_scope,verification_state).
```

The passport identifies a source artifact. It does not certify every theorem attributed to that artifact. A theorem use requires an additional claim record with statement number, hypotheses, conclusion, and dependency direction.

1.2.2 Locator grammar

The durable source address has the form

```
EGA:<chapter>.<section>.<item>
edition=<journal installment>;
pages=<journal pages>;
artifact=<digital locator>;
claim=<atomic handle>
```

For example, the generic-flatness theorem below is addressed as `EGA:IV.6.9.1`, with edition `IV_2`, journal issue 24 (1965), and the NUMDAM artifact `PMIHES_1965__24__5_0`. A bare citation such as "EGA IV" is insufficient for theorem evidence. A page number without a numbered statement is also fragile because reprints and extracted scans may use different physical pagination.

1.3 Publication form as mathematical infrastructure

The serial form affected how EGA could be read and maintained. Chapter 0, called *Préliminaires*, is not a single introductory block printed once. It continues across installments, supplying commutative algebra, homological algebra, sheaf theory, representable functors, limits, and other prerequisites near the chapters that need them. EGA III, first part, begins

with Chapter 0 §8 on representable functors before entering Chapter III. EGA IV, first part, begins with further Chapter 0 sections on dimension, depth, regular rings, formal smoothness, Cohen rings, derivations, differentials, and Japanese rings before resuming Chapter IV.

Status `S-001-CH0`; claim `CLM-V1-001-CH0`. Chapter 0 functions as a dependency waist: algebraic results are placed where the geometric development needs a stable reference, reducing the temptation to hide substantial commutative algebra inside later geometric proofs. "Dependency waist" is this chapter's design synthesis, not EGA terminology.

The publication also maintains internal reference discipline. EGA III explains that references to Chapter 0 sections published with Chapter I will be marked 0_{I}. EGA IV uses corresponding installment-sensitive symbols and includes errata lists. Such devices are evidence that the work was conceived as a living, cross-referenced system rather than a sequence of stand-alone papers.

This system has costs. A statement can depend on a preliminary result printed hundreds of pages and several years earlier. OCR can confuse Roman installment labels, zero with the letter O, superscripts, accents, and mathematical symbols. A modern citation that omits the part number may point to the wrong physical volume. The publication architecture therefore creates a provenance problem as well as solving a mathematical one.

The split of EGA III is mathematically intelligible. The first part develops cohomology of quasi-coherent and coherent sheaves, projective techniques, finiteness for proper morphisms, and formal comparison. The second part develops Tor and Künneth formalisms, cohomology and base change, homological functors, semicontinuity, and exchange. The split of EGA IV is larger: its four parts carry the local theory from foundational commutative algebra and basic local properties through flatness and fibers, limits and constructibility, dimension theory, differential invariants, smoothness, étaleness, and related local structures.

Status `S-001-SERIAL`; claim `CLM-V1-001-SERIAL`. The serial publication form mirrors a proof dependency graph only imperfectly. Earlier publication usually supplies language used later, but some sections announce future results, later parts correct or sharpen earlier statements, and certain topics are intentionally postponed. Chronological precedence is therefore evidence of availability, not a proof of logical minimality.

1.4 The four-stratum mathematical architecture

1.4.1 Stratum A: language, affine duality, and relative objects

EGA I supplies the basic object language. Affine schemes arise from prime spectra equipped with structure sheaves. General preschemes are locally affine. Morphisms are local morphisms of locally ringed spaces. Fiber products organize base change. Quasi-coherent modules globalize module theory. Reduced, integral, irreducible, separated, quasi-compact, finite-type, and related conditions become reusable geometric predicates.

The decisive feature is relativity. An S-scheme is not merely a scheme with an ignored label; it is a scheme equipped with a structural morphism to S. Morphisms between S-schemes are required to respect those structural maps. The resulting slice perspective permits the same geometric object to be studied over different bases and makes base extension a first-class construction.

Status `E-001-AFF`; theorem record `THM-V1-001-AFF`. In EGA I, Theorem 1.7.3 and Corollary 1.7.4, morphisms between affine schemes correspond contravariantly to ring homomorphisms. The source hypotheses are that both objects are affine schemes; the conclusion is an identification of the morphism set with homomorphisms between their rings in the reverse direction. This theorem does not say that every geometric question is affine.

Status `E-001-LOCAL`; definition record `DEF-V1-001-LOCAL`. EGA I, Definition 2.1.2, defines a prescheme as a ringed space in which every point has an affine open neighborhood. Proposition 2.1.3 says that affine opens form a basis. The local construction is therefore coupled to an explicit global gluing environment.

The language layer is not only a dictionary. It determines what later theorems can quantify over. Nilpotent thickenings are retained; arbitrary base schemes are allowed; points include generic and nonclosed points; local rings are attached to all points; and morphisms carry sheaf maps. This is the ambient world in which properness, smoothness, cohomology, and descent can be stated uniformly.

1.4.2 Stratum B: global morphism classes and geometric constructions

EGA II opens with a synopsis of affine morphisms, homogeneous prime spectra, relative Proj, ample sheaves, quasi-affine and quasi-projective morphisms, proper and projective morphisms, integral and finite morphisms, valuative criteria, blowups, projecting cones, and projective closure. Its introduction also says that these classes are studied with little use of cohomological methods, while a deeper study using cohomology will follow in Chapter III.

This explicit interface matters. EGA II does not pretend that global properties are independent of local algebra or cohomology. It establishes the objects and formal stability properties that cohomological theorems later consume.

Status `E-001-PROPER`; definition record `DEF-V1-001-PROPER`. EGA II, Definition 5.4.1, calls $f : X \to Y$ proper when it is separated and of finite type and when every base change $X \times_Y Y' \to Y'$ is closed. Proposition 5.4.2 records, among other facts, stability under composition and base change. The definition's universal quantifier over $Y' \to Y$ is part of the claim; replacing it by "closed" without universality changes the property.

Status `E-001-PROJ`; theorem record `THM-V1-001-PROJ`. EGA II, Definition 5.5.2 defines projective morphisms through the equivalent conditions of Proposition 5.5.1, involving closed immersion into a relative projective bundle or a relative Proj presentation. Theorem 5.5.3 states that every projective morphism is quasi-projective and proper. Its converse is stated under the condition that the base is quasi-compact or has noetherian underlying topological space: a quasi-projective proper morphism is then projective. Omitting the base hypothesis would enlarge the source statement.

The distinction between proper and projective illustrates EGA's architectural discipline. Projectivity is a presentation property with an ample line bundle or relative Proj model. Properness is intrinsic and universally closed under the stated finiteness and separation conditions. Theorem 5.5.3 relates them without identifying them unconditionally.

1.4.3 Stratum C: coherent cohomology and controlled variation

EGA III changes the proof technology. It treats higher direct images, coherent sheaves, projective morphisms, proper morphisms, formal completion, Tor, hypercohomology, semicontinuity, and base change. The layer depends on the scheme and morphism language but produces new invariants that can detect global behavior invisible to local affine calculations.

Status `E-001-FIN`; theorem record `THM-V1-001-FIN`. EGA III, first part, Theorem 3.2.1 states: if Y is locally noetherian, $f : X \to Y$ is proper, and $\mathcal{F}$ is a coherent $\mathcal{O}_X$-module, then $R^q f_*(\mathcal{F})$ is coherent for every $q \geq 0$. Each hypothesis is load-bearing. The source does not state the result for an arbitrary morphism, arbitrary base, or arbitrary $\mathcal{O}_X$-module.

Status `E-001-FORMAL`; theorem record `THM-V1-001-FORMAL`. EGA III, first part, Theorem 4.1.5 is the formal-functions comparison for a proper morphism of noetherian preschemes, a closed subset of the base, and a coherent sheaf. It identifies the relevant

completed higher direct-image objects through topological isomorphisms. The source itself describes the result as expressing, among other things, that formation of the higher direct images commutes with completion.

Status `E-001-SEMI`; theorem record `THM-V1-001-SEMI`. EGA III, second part, Theorem 7.7.5 assumes a locally noetherian base with an affine open cover, a proper morphism, and a bounded-below complex of coherent O_X-modules flat over the base. It gives semicontinuity and exchange properties for the associated homological functor. This chapter uses only the source's architectural role: coherent cohomology varies under typed finiteness and flatness hypotheses. It does not collapse the theorem into the unqualified slogan "cohomology commutes with base change."

These results demonstrate why the treatise cannot be summarized as "schemes replace varieties." The scheme language supplies the world, but EGA's force also lies in theorem families that control how global invariants behave in families. Properness, coherence, noetherian assumptions, and flatness mediate the passage from a geometric family to statements about direct images and fiberwise dimensions.

1.4.4 Stratum D: local properties, approximation, and infinitesimal tests

EGA IV develops the local study of schemes and morphisms on a much larger scale. Its first part includes an extensive continuation of Chapter 0 and begins Chapter IV. Its second part treats base change, flatness, fibers, dimension, and generic flatness. Its third part treats projective limits, constructibility, and approximation. Its fourth part develops differential invariants and systematic definitions and criteria for smooth, unramified, and étale morphisms.

Status `E-001-GFLAT`; theorem record `THM-V1-001-GFLAT`. EGA IV, second part, Theorem 6.9.1 assumes that Y is a locally noetherian integral prescheme, $u : X \to Y$ is of finite type, and $\mathcal{F}$ is a coherent O_X-module. It concludes that there is a nonempty open $U \subseteq Y$ such that $\mathcal{F}|_{u^{-1}(U)}$ is flat over U. The theorem is generic flatness in this source form. It does not assert flatness over all of Y.

Status `E-001-SMOOTHDEF`; definition record `DEF-V1-001-SMOOTHDEF`. EGA IV, fourth part, Definition 17.3.1 defines a smooth, unramified, or étale morphism as one that is locally of finite presentation and respectively formally smooth, formally unramified, or formally étale. The source notes that étale means both smooth and unramified.

Status `E-001-SMOOTH`; theorem record `THM-V1-001-SMOOTH`. EGA IV, Theorem 17.5.1 gives equivalent pointwise conditions for a locally finite-presentation morphism to be smooth at x: formal smoothness of the local algebra, and equivalently flatness at x together with geometric regularity of the fiber at x, among the listed conditions. Corollary 17.5.2 globalizes the equivalence with regular morphisms in the source's terminology.

Status `E-001-ETALE`; theorem record `THM-V1-001-ETALE`. EGA IV, Theorem 17.6.1 gives equivalent conditions for a locally finite-presentation morphism to be étale at x, including smooth plus unramified, smooth plus quasi-finite, flat plus unramified, and formal étaleness of the local algebra, with the exact residue-field condition stated in the theorem. These equivalences connect infinitesimal lifting, fiber geometry, dimension, and flatness.

The local layer does not negate the global layer. It supplies tests and openness results that let global constructions be controlled point by point and then reassembled. Conversely, local regularity alone does not produce properness or coherent direct-image finiteness. The layers interact without being reducible to one another.

1.5 The theorem spine as a typed dependency graph

The theorem spine of this historical chapter is intentionally small. Its purpose is to show how source facts support an architectural reading, not to duplicate later technical chapters.

Record	Exact EGA locator	Typed hypotheses	Conclusion used here
`THM-V1-001-AFF`	I, 1.7.3–1.7.4	affine schemes X, Y	affine morphisms correspond contravariantly to ring maps
`DEF-V1-001-LOCAL`	I, 2.1.2–2.1.3	locally ringed space	local affine neighborhoods define a prescheme; affine opens form a basis
`DEF-V1-001-PROPER`	II, 5.4.1–5.4.2	separated finite-type morphism plus all base changes	universal closedness and stability properties
`THM-V1-001-PROJ`	II, 5.5.1–5.5.3	relative projective presentation; converse has base hypothesis	projective implies proper; bounded converse
`THM-V1-001-FIN`	III, 3.2.1	Y locally noetherian; f proper; $\mathcal{F}$ coherent	$R^q f_* \mathcal{F}$ coherent
`THM-V1-001-FORMAL`	III, 4.1.5	proper morphism of noetherian preschemes; closed base subset; coherent sheaf	formal comparison for completed higher direct images
`THM-V1-001-SEMI`	III, 7.7.5	locally noetherian base; proper f; coherent base-flat bounded-below complex	semicontinuity and exchange package
`THM-V1-001-GFLAT`	IV, 6.9.1	locally noetherian integral Y; finite-type $X \to Y$; coherent sheaf	flatness after restriction to a nonempty open
`DEF-V1-001-SMOOTHDEF`	IV, 17.3.1	locally finite-presentation morphism	smooth, unramified, and étale via formal lifting properties
`THM-V1-001-SMOOTH`	IV, 17.5.1–17.5.2	locally finite-presentation morphism at a point	smoothness equivalent to flatness plus geometrically regular fiber
`THM-V1-001-ETALE`	IV, 17.6.1	locally finite-presentation morphism at a point	equivalent étaleness criteria

1.5.1 Result-local erratum dispositions

The checked correction corpus is finite and edition-specific: EGA II, "Errata et Addenda, Liste 1," pp. 217–221; EGA III, seconde partie, "Liste 2," pp. 85–91; and EGA IV, quatrième partie, "Liste 3," pp. 345–361, together with the end matter of the cited installments. A disposition of "no applicable entry found" is limited to that corpus. It does not claim that no correction exists anywhere.

Theorem-spine record	Checked disposition	Effect on the chapter use
THM-V1-001-AFF	**applicable addendum**: Liste 1 explicitly says that EGA I, 1.7.3 can be generalized and supplies the locally-ringed-space formulation beginning at 1.8.1; no retraction of I, 1.7.3 or Cor. 1.7.4 was found	retain the narrower affine-scheme anti-equivalence used here; do not cite the addendum as if it changed the original conclusion
DEF-V1-001-LOCAL	no applicable entry found for I, 2.1.2–2.1.3	retain the definition and affine-open-basis proposition
DEF-V1-001-PROPER	no applicable entry found for II, 5.4.1–5.4.2	retain the stated definition and stability use
THM-V1-001-PROJ	no applicable entry found for II, 5.5.1–5.5.3	retain the projective-implies-proper and bounded-converse scope
THM-V1-001-FIN	no applicable entry found for III, 3.2.1; the Liste 2 entry labelled 0_{I}, 3.2.1 concerns the preliminary Chapter 0 installment, not this theorem	retain the coherent-direct-image statement
THM-V1-001-FORMAL	no applicable entry found for III, 4.1.5	retain the formal-comparison statement
THM-V1-001-SEMI	no applicable entry found for III, 7.7.5	retain the semicontinuity and exchange package with its stated hypotheses
THM-V1-001-GFLAT	no applicable entry found for IV, 6.9.1	retain generic flatness on a nonempty open
DEF-V1-001-SMOOTHDEF	no applicable entry found for IV, 17.3.1	retain the lifting-property definition
THM-V1-001-SMOOTH	no applicable entry found for IV, 17.5.1–17.5.2	retain the flatness plus geometrically regular fiber criterion
THM-V1-001-ETALE	no applicable entry found for IV, 17.6.1	retain the stated equivalent criteria

Disposition CLS-V1-001-ERRATA; status S. The predecessor obligation OBL-V1-001-ERRATA is closed for the eleven records and the checked PMIHÉS correction corpus above. It must be reopened if a different edition is adopted, another correction source is admitted, or a downstream chapter uses a proof line not represented by the bounded conclusion in this table. This is a source-binding disposition, not mathematical review or admission.

The dependency arrows are typed as follows:

$$\text{affine duality} \longrightarrow \text{schemes and base change} \longrightarrow \text{morphism classes} \longrightarrow \text{cohomological and local theorems.}$$

This display encodes conceptual prerequisite order. It is not a formal proof that each theorem uses every preceding record. The actual proof dependency for an admitted theorem must be extracted from its cited proof and internal references.

Status `S-001-DEP`; claim `CLM-V1-001-DEP`. EGA's architecture is recursive rather than merely additive. Later properties are repeatedly defined by asking how objects or morphisms respond to localization, base change, completion, fibers, or infinitesimal extensions. Earlier language is therefore re-entered as a test environment for later theorems.

1.5.2 A preservation matrix

Construction or property	Composition	Base change	Locality	Required caution
affine morphism	stable in standard source results	stable	local on base in appropriate form	affine source and affine morphism are different notions
proper morphism	EGA II 5.4.2	EGA II 5.4.2	local on base	"closed" is weaker than universally closed
projective morphism	controlled by relative Proj results	stable in EGA II	presentation-dependent	proper need not be identified with projective without hypotheses
coherent direct image	functorial derived setting	exchange requires hypotheses	local on base for coherence	properness and noetherian/coherence assumptions cannot vanish
flatness	transitive in typed settings	stable	local	generic flatness yields an open locus, not global flatness
smooth or étale	stable under composition and base change	stable	local on source and target	locally finite presentation is part of the definition

The matrix is a navigation aid. Every cell intended for theorem use must resolve to an exact source statement. It must not be cited as if the table itself proved the assertions.

1.5.3 Worked trace: from properness to coherent direct images

One concrete trace shows why architectural reading is stricter than a list of famous results. Begin with EGA II, Definition 5.4.1. Properness packages three requirements: separation, finite type, and universal closedness. The universal condition is quantified over every base change $Y' \to Y$. EGA II, Proposition 5.4.2 then proves the stability needed to use proper morphisms as a class: closed immersions are proper, composites of proper morphisms are proper, and base changes of proper morphisms are proper.

EGA III, Theorem 3.2.1 consumes that class. Its input is not an informal compactness analogy but a proper morphism $f : X \to Y$, a locally noetherian base, and a coherent sheaf $\mathcal{F}$. Its output is coherence of every higher direct image $R^q f_*(\mathcal{F})$. The proof uses earlier projective machinery and a Chow-lemma reduction, so the path crosses EGA II's distinction between projective and proper morphisms rather than erasing it.

The next edge leads to EGA III, Theorem 4.1.5. Formal completion is taken along a closed part of a noetherian base, and properness plus coherence permit a comparison between

completed higher direct images and inverse systems on the infinitesimal neighborhoods. The edge

$$\text{properness} \longrightarrow \text{coherent higher direct images} \longrightarrow \text{formal comparison}$$

is therefore a sequence of theorem applications with changing ambient data. The first node is a property of a morphism. The second is a finiteness statement about derived direct images. The third is a comparison involving a chosen closed locus, induced thickenings, and topologies. Treating all three as the slogan "proper families behave well" would hide the mathematical types that make the statements usable.

Status `S-001-PROPERTRACE`; claim `CLM-V1-001-PROPERTRACE`. The trace demonstrates a general EGA pattern: define a morphism class by invariance requirements, prove closure properties, then use that class as a hypothesis for finiteness and comparison theorems. This is a source-grounded interpretation of the cited records, not an assertion that every EGA theorem has this form.

The trace also identifies losses. Properness alone does not select a coherent sheaf. Coherence of $R^q f_* \mathcal{F}$ alone does not supply a base-change isomorphism. Formal comparison does not imply algebraization without further theorems. Each step preserves some fields and introduces new hypotheses.

1.5.4 Worked trace: from finite presentation to smooth and étale maps

A second trace begins with finite-presentation control. EGA IV's approximation theory studies objects over filtered limits and explains why finite presentation is the condition under which morphisms and relations descend to a finite stage. This is categorically different from noetherianity: finite presentation constrains how an object is specified relative to its base, whereas noetherianity constrains ascending chains and finiteness behavior in the rings and spaces involved.

Definition 17.3.1 then combines local finite presentation with infinitesimal lifting. A smooth morphism is locally of finite presentation and formally smooth; an unramified morphism is locally of finite presentation and formally unramified; an étale morphism is locally of finite presentation and formally étale. The formal adjective tests extensions by nilpotent ideals. The finite-presentation adjective prevents the lifting property from ranging over objects with uncontrolled infinite presentation.

Theorem 17.5.1 translates the smoothness definition at a point. Under local finite presentation, smoothness is equivalent to flatness together with geometric regularity of the fiber, and to the listed formal-smoothness condition on the local algebra. Theorem 17.6.1 performs the corresponding translation for étaleness, connecting formal étaleness, smoothness plus unramifiedness, smoothness plus quasi-finiteness, and flatness with the exact residue-field condition.

The architecture here has three views of one phenomenon:

1. the **functorial view**, expressed by lifting against nilpotent extensions;
2. the **relative algebra view**, expressed by local finite presentation and flatness;
3. the **fiber view**, expressed by geometric regularity, dimension, and separability.

Their equivalence is theorem content. A machine record may not choose the most convenient view and delete the bridge theorem. For example, "all fibers are regular" without flatness is not the cited smoothness criterion, and "quasi-finite" without smoothness is not the cited étaleness criterion.

Status `S-001-SMOOTHTRACE`; claim `CLM-V1-001-SMOOTHTRACE`. EGA IV's smoothness and étaleness architecture connects infinitesimal, algebraic, and fiberwise tests

under finite-presentation hypotheses. The synthesis is useful for machine navigation because it offers multiple evidence routes while preserving the theorem that relates them.

This trace also explains the importance of nilpotents announced in EGA I. Infinitesimal lifting problems are posed precisely against nilpotent thickenings. Nilpotent structure is therefore not only tolerated as an enlargement of the object class; it becomes a probe for classifying morphisms. The historical introduction and the later technical architecture meet at a mathematically exact interface.

1.6 Historical fact, theorem, interpretation, synthesis, and open claim

EGA attracts compressed narratives: it "invented schemes," "completed algebraic geometry," or "replaced coordinates by categories." Such phrases may orient a lecture, but they are too coarse for source lineage. This chapter separates five logical kinds.

First, a **historical fact** concerns what a dated source says or how it was published. The eight-installment journal sequence and the thirteen-chapter plan are historical facts. They are verified by title pages and the EGA I introduction.

Second, a **theorem** has mathematical hypotheses and a conclusion. Generic flatness and proper direct-image coherence are theorems. Their historical importance does not relax their hypotheses.

Third, an **interpretation** proposes a way to understand relations among sources. The four-stratum architecture is an interpretation. It should be judged by explanatory fidelity, not confused with a numbered proposition.

Fourth, a **design synthesis** defines a new interface for GAI-MUA. The source passport, claim records, and audit algorithm below are design syntheses. EGA informs them, but they are not attributed to Grothendieck or Dieudonné.

Fifth, an **open or disputed claim** lacks enough evidence for established status. Questions about the exact private distribution of labor, the counterfactual contents of unpublished chapters, or whether a later source is the unique intended realization of an announced chapter remain Q unless a specific documentary argument is supplied.

Status `Q-001-CONTINUATION`; claim `CLM-V1-001-CONTINUATION`. It is not established here that any one later book or seminar volume is *the* canonical continuation of a projected EGA chapter. Topic overlap, authorship overlap, and cross-reference are relevant evidence, but none alone proves identity of works.

Status `Q-001-LABOR`; claim `CLM-V1-001-LABOR`. The title-page formula establishes collaboration in rédaction. It does not by itself quantify authorship, invention, checking, rewriting, or editorial responsibility at the level of individual results. This chapter does not infer those private facts.

Status `C-001-MINIMALITY`; conjecture `CONJ-V1-001-MINIMALITY`. A future formal dependency extraction may show that a substantial subset of EGA's published theorem graph factors through a comparatively small interface of relative constructions, locality principles, and preservation lemmas. This is a conjecture about a formalized dependency graph, not a mathematical claim made by EGA and not an assertion that the omitted nodes are dispensable for human understanding.

1.7 Rising-sea test: gain, cost, and return

The "rising sea" metaphor is useful only when translated into auditable questions. What problem is submerged by a more general language? What old results are recovered? What new obligations appear?

For EGA, the structural gain is substantial. Schemes permit arithmetic and geometric bases in one category. Nilpotents encode infinitesimal information. Fiber products make base change explicit. Quasi-coherent and coherent sheaves translate module theory into geometry. Relative Proj and ample sheaves organize projective constructions. Properness becomes an intrinsic universally stable condition. Cohomological finiteness and semicontinuity control variation in families. Smooth and étale morphisms connect lifting properties, flatness, and fiber regularity.

The cost is equally real. Variance becomes unavoidable. Properties acquire long lists of hypotheses. A geometric term may have absolute and relative forms. Finiteness conditions split into finite type and finite presentation. Base change can preserve an object while changing a proof obligation. Universe and size questions emerge when categories of all schemes or functors are formed. The source itself becomes a network of installments, preliminary sections, internal references, and errata.

The return to the original problem is therefore conditional. A classical projective variety can be recovered as a scheme, and familiar constructions can be expressed in the relative language. But the expanded world includes objects with nilpotents, nonclosed points, and bases beyond fields. Recovery is not identity of categories without chosen embeddings and hypotheses.

Status `S-001-RISING`; claim `CLM-V1-001-RISING`. EGA's "sea level" rises by converting exceptional techniques into instances of relative constructions and preservation theorems. The gain is not that every proof becomes short. It is that proofs can share typed infrastructure.

1.7.1 Failure boundaries

1. Affine duality does not imply that every scheme is affine.
2. A property stable under base change need not descend along an arbitrary base change.
3. Proper does not mean projective without additional hypotheses.
4. Coherent direct-image finiteness does not hold for an arbitrary morphism or arbitrary sheaf in the form cited here.
5. Generic flatness supplies a dense open locus under its source hypotheses; it does not erase the nonflat complement.
6. Smoothness includes local finite presentation; formal smoothness alone is not the cited definition.
7. Étale is not synonymous with a topological covering map.
8. A projected chapter title is not a published theorem family.
9. An exact bibliographic locator does not compensate for a misstated theorem.
10. A modern reformulation may be equivalent, stronger, weaker, or only analogous; the relation must be proved rather than presumed.

1.8 Machine interface: claim, trace, and obligation records

The machine interface is chapter-local. It is designed to prevent source history from being flattened into an undifferentiated citation string.

Status `D-001-CLAIM`; definition `DEF-V1-001-CLAIM`. An `AtomicClaimRecord` has fields

```
claim_id
chapter_id
status
claim_text
```

```
    mathematical_ambient
    hypotheses[]
    conclusion
    source_passport_id
    exact_locator
    quotation_mode
    dependencies[]
    downstream_permissions[]
    failure_boundary
    obligations[]
    claim_sha256
```

`status` is one atomic member of `E`, `S`, `D`, `R`, `O`, `C`, or `Q`. `quotation_mode` is one of `verbatim`, `faithful-translation`, `modern-reformulation`, `derived-here`, or `interpretation`. A record with status `E` and mode `modern-reformulation` must preserve the source hypotheses and declare any notation change.

Status `D-001-TRACE`; definition `DEF-V1-001-TRACE`. A `TraceEdgeRecord` has fields

```
    trace_id
    from_claim
    to_claim
    relation
    direction
    preserved_fields[]
    lost_fields[]
    required_bridge
    counterexample_or_failure
    review_state
```

Allowed relations include `textually-supports`, `defines`, `theorem-depends-on`, `reformulates`, `specializes`, `generalizes`, `historically-precedes`, `interprets`, and `does-not-imply`. The relation `historically-precedes` never grants theorem dependency.

Status `D-001-OBL`; definition `DEF-V1-001-OBL`. An `ObligationRecord` has fields

```
    obligation_id
    owner
    trigger
    required_evidence
    allowed_resolution
    blocking_scope
    independent_reviewer_required
    state
```

The state is descriptive, not self-promoting. A chapter may create or discharge source-extraction obligations in its own evidence area, but it may not grant independent mathematical review of its own claims.

1.8.1 Example atomic records

```
    claim_id: CLM-V1-001-GFLAT
    status: E
```

```
claim_text: Generic flatness holds in the exact sour
ce form stated in §1.4.4.
source_passport_id: SRC-001-EGA-IV-2-1965
exact_locator: EGA IV, Theorem 6.9.1, journal p. 163
hypotheses:
- Y locally noetherian and integral
- u:X->Y of finite type
- F coherent on X
conclusion:
- a nonempty open U of Y exists
- F restricted over U is U-flat
failure_boundary:
- no claim of flatness on all of Y
claim_id: CLM-V1-001-FOUR-STRATA
status: S
claim_text: The published EGA can be navigated throu
gh four interacting strata.
source_passport_id: null
exact_locator: this chapter §§1.4-1.5
dependencies:
- CLM-V1-001-PUBSEQ
- CLM-V1-001-PLAN13
failure_boundary:
- not exhaustive
- not an EGA theorem
claim_id: CLM-V1-001-CANONICAL-CONTINUATION
status: Q
claim_text: A unique later source is the canonical c
ontinuation of each unpublished projected chapter.
source_passport_id: null
exact_locator: unresolved
failure_boundary:
- topic overlap is insufficient
obligations:
- OBL-V1-001-CONTINUATION
```

1.8.2 Claims, traces, universes, and permissions

The claim universe is not the same as the object universe. Mathematical objects may be $\mathbb{U}$-small while the collection of source artifacts, editions, and claims is represented in a $\mathbb{V}$-small registry. A formal implementation must not quantify over "all statements" without a declared encoding universe.

Downstream permissions are monotone with respect to evidence, not status letters. An `E` bibliographic fact may authorize a publication-date statement but not a theorem. An `S` architecture may guide navigation but may not close a proof dependency. A `Q` record may motivate research but must remain outside established closure. An `O` record blocks only its declared scope.

1.9 Bounded source-audit algorithm

Status `R-001-AUDIT`; program `RP-V1-001-AUDIT`. The following bounded algorithm verifies one proposed EGA source binding. It does not search the whole literature, discover a theorem, or prove the mathematical statement.

Inputs

1. one atomic candidate claim;
2. one declared EGA installment or a finite set of candidate installments;
3. one proposed numbered locator;
4. one downstream use;
5. a finite corpus of source artifacts with hashes.

Procedure

1. Validate that the claim has one atomic status and one chapter owner.
2. Normalize the installment label, preserving Roman chapter and part numbers.
3. Resolve issue, year, page span, DOI, and digital locator.
4. Bind the consulted artifact hash.
5. Locate the numbered definition, theorem, proposition, corollary, remark, or introduction passage.
6. Extract the source hypotheses and conclusion without silently modernizing them.
7. Record notation and terminology translations separately.
8. Follow internal references needed to type the statement.
9. Check whether an erratum in the consulted corpus affects the locator.
10. Compare the downstream wording against the extracted claim.
11. Emit either a `SourceClaimCertificate` or a typed failure.

Failure codes

```
AMBIGUOUS_EDITION
MISSING_PART
UNRESOLVED_LOCATOR
ARTIFACT_HASH_MISSING
OCR_UNSUPPORTED
HYPOTHESIS_DROPPED
CONCLUSION_STRENGTHENED
BROKEN_INTERNAL_REFERENCE
ERRATUM_OPEN
STATUS_MISROUTED
SCOPE_PREEMPTION
```

A successful certificate means only that the citation binding passed the declared audit. Mathematical correctness may still require proof checking, translation review, or specialist judgment.

Status `O-001-OCR`; obligation `OBL-V1-001-OCR`. Any theorem extracted from OCR must be checked against the page image when accents, relation signs, subscripts, quantifiers, or negations could change meaning. Resolution requires a page-image comparison recorded outside the claim text.

Disposition `CLS-V1-001-ERRATA`; status `S`. Section 1.5.1 records the result-local checks and the one applicable generalizing addendum. The closed scope is the current eleven-record spine and the named PMIHÉS correction lists; an edition or claim-scope change reopens the predecessor obligation.

Status O-001-REVIEW; obligation OBL-V1-001-REVIEW. Independent mathematical and source review must verify the theorem records in §1.5. This chapter does not grant that review.

1.10 Figure notes

The figure design for this chapter comprises a mathematical, system, and algorithm triad:

- FIG-V1-CH-001-MATH, the EGA publication and theorem-family architecture;
- FIG-V1-CH-001-SYS, the typed provenance-record system;
- FIG-V1-CH-001-ALG, the bounded EGA source-audit workflow.

Status D-001-FIGBRIEF; definition DEF-V1-001-FIGBRIEF. The chapter figure note requires each figure to expose claim handles, source handles, dependency direction, failure boundaries, caption, long alt text, and exact role. The mathematical view must distinguish conceptual dependence from theorem implication. The system view must keep E, S, D, R, O, C, and Q states explicit. The algorithm view must state that a successful audit verifies source binding rather than proof.

1.11 Comparison layer: what EGA is and is not

EGA is a primary source-bearing treatise, not a software module. Its chapters contain definitions, theorem families, proof methods, examples, remarks, forward references, and preliminary algebra. A machine architecture may learn from its organization, but it must not redescribe EGA I, II, III, and IV as components with executable guarantees.

EGA is also not identical with the entire Grothendieckian source constellation. SGA's seminar form, FGA's representability and moduli developments, and later manuscripts have different publication histories and different proof-provenance structures. This chapter's architecture supplies interfaces to them:

- EGA's scheme and morphism language is prerequisite infrastructure for much of FGA and SGA.
- EGA II and III provide global and cohomological conditions used in moduli problems.
- EGA IV's flatness, finite-presentation, smoothness, and étaleness language supports later descent and étale-topology developments.
- Announced but unpublished EGA themes help explain the breadth of the original program, but do not transfer source ownership.

The relation to modern references also needs a typed bridge. A modern source may restate an EGA theorem using "scheme" where the journal says "prescheme," replace noetherian finite type by finite presentation, or separate a theorem into lemmas. Such a source can be valuable for verification and pedagogy. It cannot silently replace the primary locator when the claim is explicitly historical or edition-sensitive.

Status S-001-MODERN; claim CLM-V1-001-MODERN. A robust source record should preserve both the primary EGA address and, when available, a stable modern verification address. The two locators have different roles: the primary source supports historical attribution and original scope; the modern source may support corrected notation, proof navigation, or a generalized statement.

1.11.1 Nonimplication register

From	Does not by itself imply
publication before another installment	logical necessity for every later theorem
an announced chapter	publication or completion of that chapter
shared subject matter with SGA or FGA	identity of source works
exact issue metadata	correctness of a theorem paraphrase
a theorem name	its hypotheses
a modern generalized theorem	that EGA stated the generalized form
a machine certificate	mathematical proof

This register is part of the chapter's source-critical purpose. It prevents a history of ideas from becoming an untyped dependency graph.

1.12 Obligations and open research boundary

The chapter closes with a finite obligation set rather than a declaration of completeness.

Obligation	Required evidence	Blocking scope
`OBL-V1-001-OCR`	page-image check for theorem-critical OCR	affected claim only
`OBL-V1-001-ERRATA`	closed locally by the result-level dispositions in §1.5.1; reopen on edition or scope change	none in the current theorem spine
`OBL-V1-001-REVIEW`	independent mathematical and source review	theorem records of the chapter
`OBL-V1-001-CONTINUATION`	documentary argument for any claimed later-source identity	continuation claim
`OBL-V1-001-FORMALIZE`	formal statement and dependency extraction with universe declarations	formal dependency claims

Status `R-001-FORMALIZE`; program `RP-V1-001-FORMALIZE`. A future formalization may encode the theorem spine as a graph whose nodes carry universes, hypotheses, source locators, and proof dependencies. The bounded initial target is not all of EGA. It is the eleven records in §1.5, together with counterexamples or failure witnesses for each nonimplication used by the machine interface.

The formal target must distinguish three levels:

1. a bibliographic artifact exists;
2. an artifact contains a statement with a given address;
3. the statement is mathematically proved and correctly transported into the target formalism.

Conflating these levels would reproduce precisely the provenance error that the interface is designed to prevent.

1.13 Conclusion: architecture without mythology

The published EGA is both more bounded and more structurally powerful than its mythology. It is bounded because the journal sequence stops after EGA IV, despite the thirteen-chapter provisional plan printed in 1960. It is structurally powerful because the published installments

do not merely collect results. They build a relative language, organize global morphism classes, control coherent cohomology in families, and develop local and infinitesimal criteria that feed back into global geometry.

The publication history and the mathematics must therefore be read together. Serial installments, continuing preliminary chapters, internal cross-references, and errata are not external decoration. They are part of the evidence architecture through which definitions and theorems can be located. Conversely, issue numbers and dates do not explain the mathematics without the dependency relations among affine duality, base change, properness, coherent direct images, flatness, smoothness, and étaleness.

The durable lesson for this monograph is disciplined separation. Historical facts receive edition-specific witnesses. Theorems retain hypotheses and numbered locators. Interpretations are labeled as interpretations. Design syntheses are not projected backward into EGA. Open or disputed claims remain outside established closure. Machine records preserve these distinctions rather than compressing them into one citation field.

Under that discipline, EGA can serve as a source constellation anchor without being turned into legend or software. Its architecture becomes navigable, its publication sequence becomes auditable, and its mathematical influence can be traced without claiming that every later development was already contained in, or uniquely determined by, the published treatise.

1.14 Primary source list

1. A. Grothendieck, *Éléments de géométrie algébrique I: Le langage des schémas*, rédigés avec la collaboration de J. Dieudonné, *Publications mathématiques de l'IHÉS* 4 (1960), 5–228, DOI 10.1007/BF02684778, NUMDAM PMIHES_1960__4__5_0.
2. A. Grothendieck, *Éléments de géométrie algébrique II: Étude globale élémentaire de quelques classes de morphismes*, same collaboration formula, *Publications mathématiques de l'IHÉS* 8 (1961), 5–222, DOI 10.1007/BF02699291, NUMDAM PMIHES_1961__8__5_0.
3. A. Grothendieck, *Éléments de géométrie algébrique III: Étude cohomologique des faisceaux cohérents*, première partie, same collaboration formula, *Publications mathématiques de l'IHÉS* 11 (1961), 5–167, DOI 10.1007/BF02684273, NUMDAM PMIHES_1961__11__5_0.
4. A. Grothendieck, same title, seconde partie, *Publications mathématiques de l'IHÉS* 17 (1963), 5–91, DOI 10.1007/BF02684890, NUMDAM PMIHES_1963__17__5_0.
5. A. Grothendieck, *Éléments de géométrie algébrique IV: Étude locale des schémas et des morphismes de schémas*, première partie, same collaboration formula, *Publications mathématiques de l'IHÉS* 20 (1964), 5–259, DOI 10.1007/BF02684747, NUMDAM PMIHES_1964__20__5_0.
6. A. Grothendieck, same title, seconde partie, *Publications mathématiques de l'IHÉS* 24 (1965), 5–231, DOI 10.1007/BF02684322, NUMDAM PMIHES_1965__24__5_0.
7. A. Grothendieck, same title, troisième partie, *Publications mathématiques de l'IHÉS* 28 (1966), 5–255, DOI 10.1007/BF02684343, NUMDAM PMIHES_1966__28__5_0.
8. A. Grothendieck, same title, quatrième partie, *Publications mathématiques de l'IHÉS* 32 (1967), 5–361, DOI 10.1007/BF02732123, NUMDAM PMIHES_1967__32__5_0.